

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

**(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)**

Department of Computer Engineering



Project Report on

eVAULT: Enhanced Blockchain Verified Access and Unified Legal Records Technology

**In partial fulfillment of the Fourth Year (Semester–VII), Bachelor of Engineering
(B.E.) Degree in Computer Engineering at the University of Mumbai
Academic Year 2025-26**

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**University of Mumbai
(AY 2025-26)**

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CERTIFICATE of Approval

This is to certify that _____ of Fourth Year
Computer Engineering studying under the University of Mumbai has satisfactorily presented the
project on “*eVAULT: Enhanced Blockchain Verified Access and Unified Legal Records
Technology*” as a part of the coursework of PROJECT-I for Semester-VII under the guidance of
Prof. (Mrs.) Lifna C.S. in the year 2025-2026.

Date

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR B.E PROJECT

Learners will be to:-

Course Outcome	Description of the Course Outcome
CO 1	Do literature survey/industrial visit and identify the problem of the selected project topic.
CO2	Apply basic engineering fundamental in the domain of practical applications FOR problem identification, formulation and solution
CO 3	Attempt & Design a problem solution in a right approach to complex problems
CO 4	Cultivate the habit of working in a team
CO 5	Correlate the theoretical and experimental/simulations results and draw the proper inferences
CO 6	Demonstrate the knowledge, skills and attitudes of a professional engineer & Prepare report as per the standard guidelines.

Abstract

In today's rapidly evolving digital landscape, many sectors are transforming, yet the Indian legal system remains entrenched in outdated, paper-based practices. This reliance leads to significant delays, human errors, and potential document tampering, undermining the efficacy, transparency, and integrity of the judicial process and eroding public trust.

eVAULT (Enhanced Blockchain Verified Access and Unified Legal Records Technology) addresses these challenges by introducing a blockchain-powered legal documentation system. It enables secure creation, submission, verification, and tracking of legal records throughout their lifecycle. By decentralizing document storage using IPFS, the system ensures records remain immutable, traceable, and protected from unauthorized changes.

Utilizing Hyperledger Fabric, a permissioned blockchain framework, eVAULT logs and audits all document transactions, while the InterPlanetary File System (IPFS) provides decentralized storage for multimedia evidence, enhancing capacity and reducing costs without compromising data integrity.

Each document receives a unique digital signature through advanced cryptographic techniques, ensuring authenticity and immutability. The system also offers real-time notifications and updates to all stakeholders, including lawyers, registrars, and judges, keeping them informed about case progress.

Real-time updates and notifications keep stakeholders—lawyers, registrars, and judges—informed of case progress. eVAULT streamlines legal workflows, enhances accountability, and fosters trust by integrating blockchain security into traditional judicial processes, making legal data secure, transparent, and accessible when needed.

Ultimately, it revolutionizes case management by integrating blockchain's security features into traditional judicial workflows, ensuring sensitive legal data is secure, transparent, and accessible when needed.

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Chapter 1: Introduction

1.1 Introduction

Many industries have embraced technology to improve their operations in today's fast-paced digital landscape, yet the legal sector remains largely dependent on outdated manual record-keeping and paper-based documentation. This reliance on antiquated methods results in inefficiencies, significant delays, human errors, and an increased risk of document tampering, all of which contribute to a decline in public trust due to burdensome administrative processes and a lack of transparency.

Legal professionals manage extensive volumes of sensitive information, including court records and evidence, which are susceptible to loss, unauthorized access, and tampering. The pressing need for modernization becomes more urgent with the growing number of cases and the complexities woven into legal procedures.

The eVAULT project emerges as a transformative solution to these pressing challenges by revolutionizing the management of legal records through the power of blockchain technology. It provides a decentralized, secure, and transparent framework that effectively addresses the inefficiencies of traditional judicial systems while ensuring the integrity and immutability of legal documentation. Each transaction, from evidence submission to verification and status updates, is logged in an unalterable ledger, thereby making tampering nearly impossible and enhancing trust among all stakeholders involved.

eVAULT empowers legal professionals—such as lawyers, registrars, stamp reporters, bench clerks, and judges—by facilitating seamless communication and straightforward access to critical evidence and information. The system guarantees a fast, accurate, and tamper-proof flow of information, minimizing fraud risks and complications through its decentralized architecture, which prevents any single entity from controlling or compromising the system.

The benefits of eVAULT go beyond mere operational efficiency; secure and transparent legal documents cultivate confidence, accountability, and responsibility among all parties in the judicial process. Ultimately, this innovative solution will expedite the dispensation of justice.

1.2 Motivation

The motivation behind the eVAULT project stems from the significant challenges facing the Indian judicial system. With one of the largest populations in the world, India also has a vast number of legal cases that are often delayed due to outdated practices. Long wait times for case resolutions, complicated paper-based documentation, and a lack of transparency have resulted in one of the largest backlogs of pending cases globally.

These inefficiencies frustrate legal practitioners and erode public confidence in the system. Manual handling of documents and physical storage create risks of loss, misplacement, and manipulation of records. This environment increases the likelihood of errors or fraud, further delaying justice by keeping stakeholders uninformed about case progress.

As the population grows, the judicial system faces an increasing burden from the rising complexity and volume of legal cases. Without modernization, this burden will only escalate, hindering productivity and diminishing the quality of service provided by legal practitioners. Such inefficiencies can delay or deny justice, compromise evidence, and lead to unfair trials.

eVAULT addresses these issues through blockchain technology, which is secure, transparent, immutable, and decentralized, making it ideal for modernizing legal documentation and case management. eVAULT eliminates the inefficiencies of traditional record-keeping, prevents document tampering, and provides a streamlined, transparent platform for legal professionals and stakeholders.

By changing how legal cases are managed, eVAULT ensures the security and traceability of evidence, from document uploads to verdicts. This will enhance confidence in the judicial process, benefiting legal professionals, their clients, and the public by fostering a more transparent, accountable, and effective legal system. Ultimately, eVAULT will lead to improved legal practices in India, where digital transformation in the judiciary becomes essential for ensuring access to justice for all.

1.3 Drawback of the Existing System

The Indian judicial system continues to rely heavily on outdated manual processes and paper-based documentation for managing legal records. This dependence on traditional methods has resulted in numerous operational inefficiencies, security vulnerabilities, and systemic challenges that significantly impact the delivery of justice.

- **Manual and Paper-Based Workflow**

The current system requires case documents to be handled manually by various personnel, leading to significant delays. Physical storage demands extensive space and resources, while document retrieval is time-consuming. Paper documents are vulnerable to damage from environmental factors like fire, floods, and pests, as well as risks of misplacement or intentional tampering, compromising the integrity of legal processes.

- **Lack of Transparency and Traceability**

The existing system provides inadequate control over document access, leading to concerns about unauthorized access and data leaks. There is no reliable mechanism to trace the flow of documents across various authorities, making accountability nearly impossible. The opacity in verification processes makes it difficult to verify document authenticity, leading to distrust among stakeholders.

- **Inconsistent Verification Process**

Document verification is entirely manual, handled by personnel like Registrars and Stamp Reporters. This makes it highly susceptible to human errors, inconsistencies, and potential bias. The time-consuming nature of manual validation increases inefficiencies and the likelihood of missing critical details. Different personnel may apply varying standards, leading to inconsistencies in document approval.

- **Biased Case Allocation**

Case files are often manually allocated to judicial personnel, leading to potential biases, favoritism, or uneven workload distribution. This results in some officials being overburdened while others remain underutilized, affecting both fairness and processing speed.

- **Absence of Real-Time Updates**

Lawyers and legal professionals are frequently left uninformed about case progress. Updates regarding case status, evidence submissions, and hearing schedules are not communicated in real-time, leading to inefficiencies and delays. Manual status updates by Bench Clerks mean changes take considerable time to reflect in the system.

- **Poor Data Integrity and Security**

The system lacks robust cryptographic mechanisms to ensure document integrity. Centralized data storage creates a single point of failure where security breaches, data corruption, or system crashes can compromise all legal records. Weak authentication mechanisms make it difficult to verify the identity of individuals accessing documents, increasing fraud risks.

- **Incomplete Filings and Audit Trails**

The current judicial process often faces issues like missing or fake stamps and improperly arranged papers, causing rejections and long delays. Additionally, the lack of a complete audit trail makes it difficult to track who accessed or modified documents, reducing transparency and accountability in case management.

- **Scalability and Performance Issues**

As case volumes and digital evidence grow, system performance degrades significantly. The infrastructure struggles to handle increasing workloads efficiently, resulting in system slowdowns, crashes, and extended downtimes that affect case processing..

- **Over-Reliance on Hard Copies**

Despite digitization initiatives, legal procedures continue to require physical paper files for official records. This perpetuates paper-based system problems and creates duplication of effort when maintaining both physical and digital copies.

- **Poor User Experience**

Where digital systems exist, they are often not user-friendly and require expert technical knowledge. Limited remote access and mobile accessibility restrict legal professionals' ability to work efficiently, especially in remote or underserved areas.

1.4 Problem Definition

➤ Problem Statement

The judicial system in India, much like in many other countries, faces several challenges in managing legal documents efficiently. Currently, case records and supporting evidence such as written documents, images, videos, and audios are handled manually and stored in physical formats, making the entire process cumbersome, prone to errors, and lacking transparency. Despite the increasing digitalization of other sectors, the legal field has been slow in adopting digital innovations to manage critical legal documents securely and effectively. The eVAULT system aims to address the gaps present in the current system by introducing blockchain technology into the legal document management process. Below is a detailed breakdown of the issues currently plaguing the system and the objectives of the proposed solution.

Current Challenges in Legal Document Management

1. Manual Process and Paper-Based Workflow:

- **Inefficiency and Time Delays:** One of the key challenges of the existing system is that case documents are handled and processed manually by various personnel, leading to significant delays. Legal professionals and court officials have to rely on physical paper documents, which are passed around for verification and processing, contributing to inefficiencies.
- **Physical Storage:** Physical storage of these documents requires extensive space and resources, and retrieving files often takes substantial time, causing further delays in judicial proceedings.
- **Vulnerability to Damage or Loss:** Paper documents are vulnerable to damage from environmental factors like fire, floods, and pests, as well as risks of misplacement or intentional tampering, which can compromise the integrity of the legal process.

2. **Lack of Transparency and Traceability:**

- **Limited Access Control:** The current system provides limited control over who can access the documents at different stages of verification, leading to concerns about data leaks or unauthorised access.
- **No Clear Audit Trail:** In the existing system, tracking the journey of case files and understanding who has accessed or modified them is difficult. There is no reliable mechanism to trace the flow of documents across various authorities in the judicial chain.

3. **Inconsistent Verification Process:**

- **Human-Dependent Verification:** Document verification is a fully manual process handled by personnel like Desk Babu (Registrar) and Stamp Reporter. The manual nature of this task means that it is susceptible to errors, inconsistencies, and, in some cases, bias. The absence of an automated way to streamline this process adds to the inefficiency.

4. **Case Allocation Issues:**

- **Non-Random Allocation:** Currently, documents are often manually allocated to judicial personnel for verification, leading to potential biases or uneven workload distribution. The lack of a structured allocation mechanism affects the fairness and speed of the verification process.

5. **Lack of Real-Time Updates:**

- **Delayed Communication:** Lawyers and legal professionals are often left in the dark when it comes to case progress. Updates on case status, new evidence, and decisions are not communicated in real-time, leading to inefficiencies and unnecessary delays in proceedings.
- **Manual Status Updates:** The manual nature of updating case statuses by Bench Clerks means that any changes to court schedules, case developments, or rescheduled dates often take time to reflect in the system, leaving parties uninformed.

➤ Objectives

To address these challenges, the eVAULT system is designed to introduce an automated, secure, and transparent method of managing legal documents using blockchain technology. The proposed system leverages Hyperledger Fabric, IPFS (InterPlanetary File System), and smart contracts to build a decentralised and tamper-proof legal document management system, while still allowing human verification. The following objectives aim to bridge the existing gaps in the legal document management process:

1. Automation of Case Allocation and File Handling:

- **Objective:** To eliminate manual bias and uneven workload distribution by automating the random allocation of case files to judicial personnel, such as the Desk Babu and Stamp Reporter.
- **Sub-Objective:** Develop an algorithm that automates case allocation, linking new verifications to previously verified documents through a unique Case ID, ensuring fairness and minimizing redundant checks.

2. Decentralized Document Storage and Traceability:

- **Objective:** To create a decentralized, immutable repository for all legal documents, ensuring secure storage, transparency, and full traceability of all access and modifications.
- **Sub-Objective:** Integrate IPFS for decentralized evidence storage and use blockchain to store metadata and document hashes, creating an immutable ledger for secure tracking of all document actions.

3. Secure Digital Verification and Signature Process:

- **Objective:** To enhance the manual verification process digitally, reducing errors and improving traceability.
- **Sub-Objective:** Implement a digital signature system where verified documents are signed by the Stamp Reporter and linked to the unique Case ID on the blockchain, ensuring tamper-proof records.

4. Real-Time Case Updates and Notifications:

- **Objective:** To provide real-time updates on case statuses, ensuring legal professionals have instant access to the latest information.
- **Sub-Objective:** Use event-driven programming to trigger real-time notifications for case status changes, ensuring prompt updates and reducing communication delays.

5. Enhanced Data Security and Privacy:

- **Objective:** To protect sensitive legal documents with encryption and access control, allowing only authorized personnel to view or modify case files.
- **Sub-Objective:** Apply role-based access control via Hyperledger Fabric, with cryptographic hashing and encryption to ensure data security and integrity.

6. Integration of Traditional Judicial Processes with Digital Systems:

- **Objective:** To integrate traditional legal processes with digital technologies for improved efficiency.
- **Sub-Objective:** Design a system that supports manual verification while enhancing efficiency with digital file handling and compliance with existing legal frameworks.

1.5 Relevance of the Project

The eVAULT project addresses critical challenges faced by the modern judicial and legal documentation ecosystem, where inefficiency, manual verification, and lack of transparency often delay justice and increase operational costs. The system is designed to revolutionize document verification and case management by leveraging blockchain and decentralized storage technologies to create a tamper-proof, transparent, and secure digital legal infrastructure.

The relevance of this project lies in its potential to digitally transform judicial workflows through automation, traceability, and accountability. It reduces dependence on physical records and manual signatures by introducing smart contracts that autonomously manage approvals, verifications, and updates. By using Hyperledger Fabric and IPFS, eVAULT ensures that every transaction and document is verifiable, immutable, and accessible only to authorized personnel.

The system also promotes collaboration across multiple judicial entities, ensuring that lawyers, registrars, stamp reporters, bench clerks, and judges operate within a single, synchronized environment. Its integration of blockchain-based auditing enhances legal transparency and public trust in digital judicial systems.

In the broader context, eVAULT aligns with government initiatives for Digital India and e-Governance, providing a scalable model for digital court systems and document verification frameworks across states and institutions.

1.6 Methodology Used

The methodology adopted for the development of eVAULT (Enhanced Blockchain Verified Access and Unified Legal Records Technology) is a structured, modular, and iterative approach that integrates blockchain technology with decentralized storage and smart contract automation. The primary goal of the methodology is to ensure security, transparency, and immutability in managing legal documentation and workflows within the judicial ecosystem.

The project development follows a systematic process that includes the following stages:

- **Blockchain Implementation using Hyperledger Fabric**

The core of the eVAULT system is implemented using Hyperledger Fabric, a permissioned blockchain framework specifically designed for enterprise applications. It ensures secure and tamper-proof document management across multiple judicial stakeholders. Each transaction, such as document uploads or case updates, is converted into a cryptographic hash and recorded immutably on the blockchain ledger.

Hyperledger Fabric's Membership Service Provider (MSP) authenticates participants and provides role-based access control, ensuring that only authorized users such as lawyers, registrars, stamp reporters, bench clerks, and judges can perform specific operations within the system.

- **Integration of IPFS for Decentralized Storage**

The InterPlanetary File System (IPFS) is integrated to provide decentralized storage for large multimedia files such as legal documents, audio-visual evidence, and scanned records. Instead of storing entire documents directly on the blockchain, only the metadata and IPFS hash (CID) are recorded.

This hybrid architecture ensures data integrity, cost efficiency, and scalability by reducing the load on the blockchain while maintaining the traceability and authenticity of every document uploaded to the network.

- **Automation through Smart Contracts**

Smart contracts form the backbone of workflow automation in eVAULT. They are deployed to automate key judicial processes, including case registration, document verification, and evidence validation. Once deployed, these contracts autonomously execute specific operations when predefined conditions are met, ensuring accuracy and eliminating human intervention.

Every transaction executed by a smart contract is permanently logged on the blockchain, enabling complete traceability and accountability of the process.

- **Event-Driven Programming for Real-Time Communication**

To improve responsiveness and collaboration among users, eVAULT employs event-driven programming. This mechanism ensures that all stakeholders receive instant notifications about key actions such as new case creation, evidence submission, or judgment updates.

Event listeners constantly monitor system events and trigger corresponding actions, ensuring a real-time and synchronized communication channel between all authorized participants within the judicial workflow.

- **Random Allocation Algorithm for Fair Case Distribution**

The system incorporates a Random Allocation Algorithm to distribute cases fairly among judicial personnel such as Registrars and Stamp Reporters. This prevents biased task allocation and promotes transparency and balanced workloads.

The algorithm dynamically analyzes the current task distribution and automatically assigns new cases to available personnel, ensuring equal participation and improved efficiency across all organizational units.

- **Testing and Validation Phase**

Once development was completed, the system underwent a rigorous testing and validation phase. Each module—Lawyer, Registrar, Stamp Reporter, Bench Clerk, and Judge—was tested individually to ensure smooth interaction and compliance with the predefined workflow.

Network validation was carried out by executing peer channel commands in the Hyperledger Fabric CLI to confirm that all organizational peers successfully joined their respective communication channels. Additionally, functionality testing verified data consistency, blockchain connectivity, and proper notification delivery across modules.

Chapter 2 Literature Survey

2.1 Survey of Existing Systems

Year	Objective	Data Input	Methodology	Performance Metrics	Advantages	Disadvantages
2024 [1]	To develop a secure and efficient legal documentation management system using Blockchain and IPFS to ensure integrity and de-centralized storage of legal documents.	Legal documents (contracts, agreements, court filings), user access data, cryptographic hashes, and smart contracts.	-Upload documents to IPFS. -Generate cryptographic hash. -Store hash on Blockchain. -Implement access control. -Verify document integrity via Blockchain. -Use smart contracts for legal processes.	Security, efficiency, scalability, and integrity in document access, storage, retrieval, and maintenance.	Enhanced security with Blockchain and IPFS, decentralized storage, document immutability, and automation through smart contracts.	High Cost, Complexity, Scalability Issues, Data Privacy Concerns.
2024 [2]	Design a smart contract architecture for automating insurance processes.	Policyholder information, insurance policy details, and claim information.	- Blockchain: For safe transaction recording. - BPM: For automating workflows. - Smart Contracts: For policy management.	Automation, Processing Time, Cost Efficiency, Fraud Prevention.	Automation, Transparency, Cost Reduction, Fraud Prevention.	Interoperability, Regulatory Compliance, Adoption Barriers.
2024 [3]	Explore blockchain for preserving digital evidence in judicial processes.	Digital Evidence (Documents, Images, Videos, Metadata).	- Blockchain: For tamper-proof evidence storage. - Digital Signatures: For authenticity verification. - Cryptographic Techniques: For securing evidence	Evidence security, retrieval efficiency, and legal admissibility.	Integrity, transparency, and trustworthiness.	Cost, complexity, and scalability.

Year	Objective	Data Input	Methodology	Performance Metrics	Advantages	Disadvantages
2024 [4]	Create a blockchain-based e-vault for securely storing legal records.	Legal Records (Case Files, Contracts, Agreements).	- Blockchain: For safe storage of records. - Smart Contracts: For document access. - Cryptographic Hashing: For document integrity.	Security of Document Storage, Accessibility, and Retrieval Efficiency.	Security, Transparency, Efficiency, and Immutability.	Implementation Cost, Technical Complexity, and Adoption Barriers.
2024 [5]	Enhance privacy in cyber-physical systems via blockchain-assisted data sharing.	Data from Cyber-Physical Systems and Encrypted User Data.	- Blockchain: For tamper-proof evidence storage. - Smart Contracts: To automate verification processes. - Cryptographic Hashing: To ensure evidence integrity.	Data Privacy, System Efficiency, and Deniability.	Enhanced Privacy, Data Control, and Legal Compliance.	Implementation Complexity and Overhead in Data Management.
2024 [6]	Examine blockchain's application in preserving electronic evidence in copyright cases.	Copyright Data, Electronic Evidence, and Legal Metadata.	-Blockchain: For secure digital evidence storage. -Digital Signatures: To verify authenticity. - Cryptographic Hashing: To maintain evidence integrity.	Evidence Integrity, Legal Admissibility, and Efficiency.	Improved Evidence Integrity, Reduced Tampering, and Trustworthiness.	Implementation Complexity and Legal and Regulatory Challenges.
2024 [4]	Create a blockchain-based e-vault for securely storing legal records.	Legal Records (Case Files, Contracts, Agreements).	- Blockchain: For safe storage of records. - Smart Contracts: For document access. - Cryptographic Hashing: For document integrity.	Security of Document Storage, Accessibility, and Retrieval Efficiency.	Security, Transparency, Efficiency, and Immutability.	Implementation Cost, Technical Complexity, and Adoption Barriers.
2024 [5]	Enhance privacy in cyber-physical systems via blockchain-assisted data sharing.	Data from Cyber-Physical Systems and Encrypted User Data.	- Blockchain: For tamper-proof evidence storage. - Smart Contracts: To automate verification processes.	Data Privacy, System Efficiency, and Deniability.	Enhanced Privacy, Data Control, and Legal Compliance.	Implementation Complexity and Overhead in Data Management.

Year	Objective	Data Input	Methodology	Performance Metrics	Advantages	Disadvantages
2024 [6]	Examine blockchain's application in preserving electronic evidence in copyright cases.	Copyright Data, Electronic Evidence, and Legal Metadata.	-Blockchain: For secure digital evidence storage. -Digital Signatures: To verify authenticity. - Cryptographic Hashing: To maintain evidence integrity.	Evidence Integrity, Legal Admissibility, and Efficiency.	Improved Evidence Integrity, Reduced Tampering, and Trustworthiness.	Implementation Complexity and Legal and Regulatory Challenges.
2024 [7]	Streamline the process of filing FIRs using a blockchain-based system to ensure secure, immutable, and transparent records, enhancing accountability.	FIR Details (incident descriptions, personal details) and Blockchain Records.	- Use Blockchain for recording FIRs. - Implement smart contracts for FIR submission. - Develop a user-friendly interface. - Utilize cryptographic hashing for integrity verification.	Transparency (visibility of FIRs), Security (prevention of unauthorized modifications).	Improved trust and transparency, and reduced delays in FIR registration.	Resistance from traditional law enforcement and challenges in ensuring accessibility for non-technical users.
2024 [8]	Propose an IoT-driven blockchain architecture for managing the healthcare supply chain and protecting medical records, addressing challenges of fragmented healthcare systems to enhance patient safety.	Medical Records, Supply Chain Data, and IoT Sensor Data.	- Design - Access Control - Consensus - Evaluation	Execution Time, Resource Consumption, and Throughput.	Enhanced Security, Improved Data Integrity, Access Control, and Traceability.	Compliance Issues, Scalability Concerns, and Integration Challenges.

Year	Objective	Data Input	Methodology	Performance Metrics	Advantages	Disadvantages
2024 [9]	The paper aims to develop a blockchain solution to reduce corruption by creating a document verification process that minimizes third-party access, ensuring document safety and integrity.	Document verification process, third-party access to sensitive information, and user authentication.	The methodology involves creating, storing, retrieving, and extracting documents using blockchain technology to eliminate third parties from the verification process, ensuring security and tamper-proof information.	The system can significantly support achieving Sustainable Development Goal 16.	The proposed solution eliminates third parties from document verification, reduces corruption, and frees up state resources.	Scalability, Adoption, and Regulatory Challenges.
2023 [10]	Develop a blockchain-based system for secure certificate verification.	Certificate Information (ID, Issuance Date, Issuing Authority).	- Blockchain: For secure, unchangeable ledger. - Smart Contracts: To automate issuance and verification. - Cryptographic Hashing: For data security.	Efficiency, Verification Time, Security, and Fraud Detection.	Security, Transparency, Efficiency, and Immutability.	Scalability, Complexity, and Integration Challenges.
2023 [11]	Conduct a comprehensive review of blockchain's role in document authentication.	Document Details, Authentication Records, and User Credentials	- Blockchain: For decentralized and secure document storage. - Encryption: For protecting document privacy. - Decentralized Storage: To ensure accessibility and control.	Accuracy, Security, Immutability, and Transparency.	Enhanced Security, Improved Trust, and Decentralization.	Integration Challenges, High Implementation Costs, and Scalability Issues.

Year	Objective	Data Input	Methodology	Performance Metrics	Advantages	Disadvantages
2023 [12]	Present an open-source blockchain application designed to modernize and secure the management of court data.	User Registration Information, Case Details, Evidence Files, and Access Credentials.	- Secure user registration with unique IDs and private keys for judges and lawyers. - Case files are accessed via blockchain with controlled access. - Cryptographic alerts are triggered on unauthorized attempts. - Sensitive information is protected, ensuring the integrity and security of court records.	The system offers secure, real-time tracking of case files using cryptographic algorithms and distributed storage, ensuring authorized access and minimizing data tampering.	Enhanced Security, Decentralization, Improved Transparency, Real-time Updates, Access Control, User-Friendly Interface, Audit Trail, and Cost Efficiency.	Resistance to Change, Legal and Regulatory Challenges, Scalability Concerns, and Privacy Issues.
2022 [13]	Survey blockchain technology architecture, consensus algorithms, smart contracts, applications, and security measures. Guide users and researchers in understanding and addressing security challenges.	Existing Literature and Case Studies.	- Review: Analyze literature on blockchain. - Comparison: Assess consensus algorithms and security measures. - Risk Analysis: Identify and categorize security risks.	Effectiveness of Consensus Algorithms and Security Measures.	Comprehensive Coverage, Detailed Comparisons, and Identification of Research Gaps.	Scope of Survey and Dynamic Nature of Technology.
2022 [14]	Create a decentralized personal document locker using blockchain.	User Documents, Metadata, and Access Control Information.	- Blockchain: For secure and transparent data transactions. - Deniable Encryption: For protecting user privacy.	Security, Accessibility, Data Integrity, and Privacy.	Decentralization, Security, Privacy, and User Control.	Usability Challenges, Data Management Complexity, and Cost of Implementation.

Year	Objective	Data Input	Methodology	Performance Metrics	Advantages	Disadvantages
2022 [15]	Create a decentralized digital document management system using blockchain and P2P networks to enhance security, accessibility, and collaboration.	Documents (digital files) and Blockchain Records (document transactions and user interactions).	- Use Blockchain for immutable document ledger. - Utilize P2P network for decentralized sharing. - Implement smart contracts for access control. - Apply cryptographic techniques for security.	Collaboration Efficiency (real-time sharing and editing), Security (resistance to unauthorized access and tampering), and Scalability (handling users and documents).	Enhanced security and privacy through decentralized control, and improved collaboration across stakeholders.	Challenges in integration with existing systems and performance issues in low-bandwidth environments.
2022 [16]	To survey and provide an in-depth analysis of blockchain technology, focusing on its potential applications, security features, and challenges.	Secondary data from existing literature, including research papers, articles, and case studies on blockchain technology.	Survey methodology, systematically reviewing existing literature on blockchain technology, covering consensus algorithms, cryptographic techniques, and real-world applications.	Throughput, Latency, Security Metrics.	Security, Transparency, Decentralization .	Scalability, Energy Consumption: High energy usage in PoW-based blockchains, and Regulatory Issues: Lack of regulation and standardization.
2022 [17]	The research aims to develop a blockchain-based evidence transfer model for China's judicial system, ensuring integrity, consistency, and non-repudiation of records.	Evidence transfer records and transaction proposals from authorized entities detailing evidence management.	System Design: Consortium blockchain for secure judicial transactions, Consensus Mechanism: Raft algorithm for tamper-proof records, Smart Contracts: Automates evidence management, Prototype	Throughput: Number of transactions per second, Latency: Time from initiation to confirmation, and Resource Overhead: Computational and storage resources consumed.	Tamper-Proofing, Decentralization , Enhanced Security, Efficiency.	Byzantine Vulnerability, Denial of Service Attacks, Resource Consumption.

Year	Objective	Data Input	Methodology	Performance Metrics	Advantages	Disadvantages
2021 [18]	Develop a secure digital locker system using BigchainDB and IPFS to store and manage digital documents, providing a decentralized, tamper-proof solution for data integrity, security, and easy access.	Documents (e.g., legal papers, certificates) and Metadata (cryptographic hashes stored on the blockchain).	- Use BigchainDB to store metadata. - Store documents on IPFS. - Generate cryptographic hash. - Implement access control. - Integrate BigchainDB and IPFS for decentralized storage.	Data Integrity, Storage Efficiency, Access Time.	Enhanced Security and Data Integrity, Decentralized Storage, Reduced Storage Redundancy.	Performance impacted by IPFS network latency, Complexity in implementation.

Table 1: Literature Survey

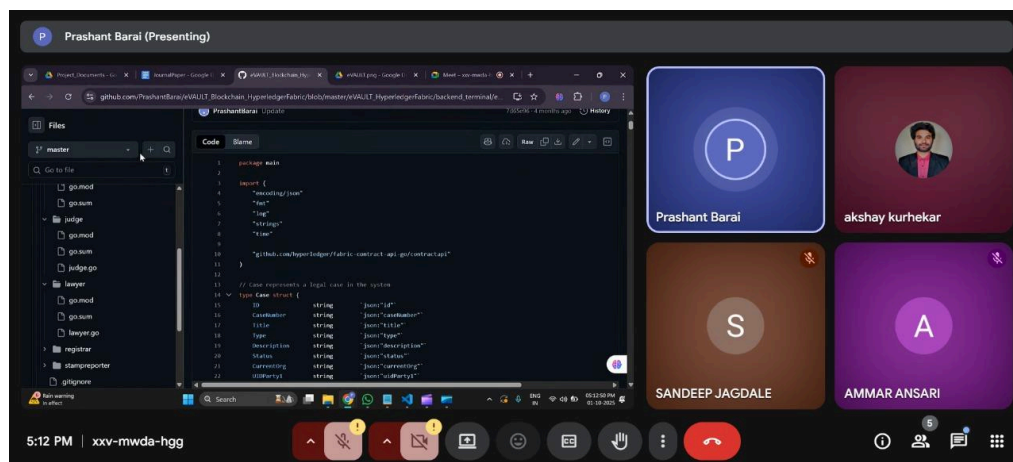
2.2 Interaction with domain experts

As part of the research and development phase of the eVAULT project, a meeting was conducted with a domain expert Akshay Kurhekar in Blockchain technology to gain a deeper understanding of blockchain network integration and its application in real-world use cases. The objective of this interaction was to receive professional guidance on optimizing blockchain deployment for secure and efficient legal documentation management.

During the session, the expert emphasized the importance of network topology design, chaincode optimization, and secure peer communication within a Hyperledger Fabric environment. To help the team better understand these technical concepts, the expert recommended studying his existing project — a Blockchain-based Electronic Health Record (EHR) Management System. This reference project demonstrated how sensitive health data can be securely stored, shared, and accessed using blockchain and smart contracts, serving as an excellent analogy for handling legal documents in eVAULT.

By exploring the EHR project, the team gained valuable insights into data privacy, access control, encryption mechanisms, and off-chain storage techniques, which are equally relevant to the legal domain. After reviewing the project architecture and implementation, the team planned a follow-up discussion to clarify specific integration-related doubts — particularly regarding multi-organization network configuration, chaincode deployment practices, and real-time event handling across blockchain peers.

This interaction not only strengthened the team's conceptual understanding of blockchain integration but also provided a practical perspective on system scalability and security, enabling the team to refine the architectural design of eVAULT accordingly.



A detailed interaction was conducted with a domain expert — Professor from a Law College, who provided valuable insights into the current legal framework and its limitations regarding digital transformation



The expert advised the team to observe live court proceedings and interact with practicing lawyers to understand the procedural flow of evidence submission and verification. He highlighted that evidence is still largely presented in physical form and digital submissions are not yet accepted under the current legal framework. This poses a challenge for projects like eVAULT that propose a fully digital workflow.

He further emphasized that critical evidence such as postmortem reports and forensic findings are handled only by government entities like hospitals and forensic labs. Therefore, building trust and collaboration with these institutions is crucial. Our system must ensure data authenticity and compliance with legal norms to gain the cooperation of such authorities.

To address these points, eVAULT aims to:

- Introduce a **hybrid workflow**, where physical evidence can be digitally mirrored and verified using blockchain-backed metadata.
- Work towards **integration with government-verified sources**, ensuring only authenticated entities can upload or validate evidence.
- Incorporate **digital notarization and certification mechanisms** to bridge the gap between physical and digital evidence systems.

This expert feedback will guide the refinement of eVAULT's design to align with real-world judicial procedures and future legal digitalization.

Chapter 3: Requirement of Proposed System

3.1 Functional Requirements

Functional requirements define the specific operations and behaviors that the eVAULT system must perform to achieve its intended objectives. These requirements ensure that all stakeholders — including lawyers, registrars, stamp reporters, bench clerks, and judges — can securely manage, verify, and track legal documents through a unified blockchain-based framework.

Case Creation and Management:

Enable lawyers to create and register new cases, upload related legal documents, and associate supporting evidence files with each case record.

Document Verification:

Provide a mechanism for registrars and stamp reporters to validate, digitally sign, and approve documents through smart contract-based verification processes.

Decentralized Evidence Storage:

Integrate IPFS (InterPlanetary File System) for decentralized storage of case files, evidence, and multimedia content, ensuring immutability and secure retrieval.

Blockchain-Based Record Maintenance:

Log every transaction — from document submission to judgment issuance — on a permissioned Hyperledger Fabric blockchain, ensuring complete traceability and data integrity.

Random Case Allocation:

Automatically allocate cases to registrars and stamp reporters using a random allocation algorithm to ensure fair workload distribution and eliminate manual bias.

Real-Time Notifications:

Provide instant notifications to stakeholders about case updates, verifications, approvals, or judgments using event-driven programming and automated triggers.

Role-Based Access Control:

Implement role-specific access rights for all judicial entities to ensure secure handling of sensitive legal information and prevent unauthorized data exposure.

Audit Trail and Case Tracking:

Maintain a complete audit history of all activities within a case, allowing authorized personnel to view the chronological flow of actions and decisions.

User Dashboard and Interface:

Offer a responsive, user-friendly dashboard for each stakeholder group, displaying case details, progress status, notifications, and evidence history in real time.

Integration with Traditional Judicial Workflow:

Ensure that the system complements existing legal verification procedures while enhancing efficiency and transparency through digital mechanisms.

3.2 Non-Functional Requirements

Non-functional requirements (NFRs) define the quality attributes, constraints, and performance expectations of the eVAULT system. These requirements determine how the system should operate rather than what specific tasks it performs. They ensure that the deployed blockchain-based legal documentation platform is secure, scalable, and efficient enough for real-world judicial use.

Performance

- **Response Time:** All user interactions, such as document upload, verification, or record retrieval, should be processed and acknowledged within 5 seconds under normal network conditions.
- **Throughput:** The system should handle multiple concurrent transactions from different roles (lawyers, registrars, and judges) without performance degradation.

- **Scalability:** The platform must support an expanding number of users and transactions, with the ability to integrate additional nodes and organizations into the Hyperledger Fabric network as required.
- **Reliability:** Ensure reliable peer communication, block generation, and synchronization between all nodes in the blockchain network to prevent transaction loss or duplication.

Security

- **Data Integrity:** All uploaded and verified documents should be cryptographically hashed before storage, ensuring that no tampering or modification occurs after upload.
- **Confidentiality:** Implement role-based access control (RBAC) so that only authorized users can view or modify data relevant to their assigned roles.
- **Authentication and Authorization:** Use digital certificates and MSP (Membership Service Provider) authentication from Hyperledger Fabric to verify participant identity and ensure trust among nodes.
- **Data Encryption:** All data transmissions between front-end, blockchain nodes, and IPFS should occur over encrypted channels using HTTPS/TLS protocols.

Usability

- **User Interface Simplicity:** Provide a clean, intuitive dashboard for all stakeholders with minimal navigation steps required for case creation, document upload, and verification.
- **Accessibility:** Ensure compatibility across major browsers and devices, maintaining full functionality on both desktop and tablet interfaces.
- **Error Handling:** Display clear, user-friendly error messages for issues like network disconnection, invalid document formats, or failed smart-contract execution.

Availability and Maintainability

- **System Uptime:** Maintain at least **99.5% availability** during operational hours, ensuring continuous blockchain service and peer accessibility.

- **Maintainability:** Use a **modular architecture** to allow independent updates of front-end components, smart contracts, or peer organizations without affecting overall system functionality.
- **Data Backup and Recovery:** Ensure periodic backups of blockchain ledgers and IPFS pin sets to allow disaster recovery and business continuity in case of node failure.

Portability and Compatibility

- **Platform Independence:** The system should be deployable across multiple operating systems such as Windows, Linux, or macOS, provided Docker and Node.js environments are available.
- **Interoperability:** Allow integration with external digital signature services, case-management tools, or e-governance APIs for future scalability and functionality expansion.

3.3 Constraints

Constraints are the technical, operational, and legal limitations that influence the design, development, and deployment of the eVAULT system. These factors must be considered carefully to ensure the project's success within real-world judicial and infrastructural environments.

Blockchain Complexity

The implementation of Hyperledger Fabric requires setting up multiple organizations, peers, orderers, and channels. This adds configuration complexity and demands technical expertise in certificate management, chaincode deployment, and Docker-based container orchestration.

Data Storage and Network Latency

Since large documents and evidence files are stored on IPFS, retrieval time can vary depending on network conditions and node availability. This may cause slight delays during document verification or evidence loading.

Integration with Existing Judicial Systems

The current legal ecosystem operates largely on traditional file-based or semi-digital processes. Integrating blockchain technology into such environments requires careful adaptation to existing policies and regulatory workflows.

Limited Computational Resources

Running multiple blockchain peers, orderer nodes, and IPFS daemons simultaneously requires significant CPU and memory resources. Low-end systems may struggle to maintain smooth operation during high transaction loads.

Legal and Compliance Constraints

The use of decentralized technologies in judicial applications must comply with data protection laws and government regulations regarding the storage, transmission, and verification of sensitive documents.

User Training and Adoption

Judicial personnel may require additional training to effectively use blockchain dashboards, understand digital signing procedures, and interpret blockchain transaction logs.

Scalability Concerns

While Hyperledger Fabric is modular, scaling the system to national-level use would require substantial infrastructure, including high availability nodes, redundant storage, and optimized consensus mechanisms.

3.4 Hardware & Software Requirements

The following hardware and software specifications define the minimum and recommended configurations for developing, deploying, and testing the eVAULT system.

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 or equivalent	Intel Core i7 / AMD Ryzen 7
RAM	8 GB	16 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
GPU (Optional)	NVIDIA CUDA-enabled GPU	For faster cryptographic operations & simulations
Network	Stable internet connection	50 Mbps or higher bandwidth

Table 2

The development and testing environment for Hyperledger Fabric and IPFS requires significant memory and processing power, especially when running multiple peers and Docker containers simultaneously.

Software Requirements

Category	Tool / Technology	Purpose / Description
Operating System	Windows 10 / Ubuntu 22.04 LTS	Development and deployment environment
Blockchain Framework	Hyperledger Fabric (v2.5+)	Core blockchain framework for distributed ledger and smart contracts
Smart Contract Language	Node.js / GoLang	Development of chaincode logic and automation scripts
Decentralized Storage	IPFS	Storage of legal documents and evidence in a decentralized network

Database	CouchDB	World state database for Hyperledger Fabric peers
Backend	Node.js	Server-side scripting and API development
Frontend	Vite + HTML/CSS/JavaScript	User dashboards and system interface
Containerization	Docker, Docker Compose	Deployment of blockchain nodes and services
Development Environment	Visual Studio Code	IDE for coding, debugging, and testing
Version Control	Git, GitHub	Source code management and collaboration
Testing Tools	Postman, Mocha	API and module testing
Documentation Tools	Microsoft Word, Draw.io	Report preparation and diagram design

Table 3

3.5 Techniques Utilized Till Date for the Proposed System

The development of the eVAULT system has so far focused on implementing key techniques that enable secure, decentralized, and automated handling of legal documentation using blockchain and distributed storage technologies. The following subsections summarize the primary techniques applied during the design and development phases.

1. Blockchain Network Configuration

The system utilizes Hyperledger Fabric as the foundational blockchain framework.

- Multiple organizations were created, each representing judicial entities such as Lawyer, Registrar, Stamp Reporter, Bench Clerk, and Judge.
- Orderer nodes were deployed to manage transaction sequencing and ensure network consensus.

- **Channel creation and joining** were configured to isolate transactions and enable private data sharing among specific peer organizations.
- **Chaincode (smart contract)** was developed and deployed to automate core judicial workflows such as document verification, case allocation, and judgment logging.

2. IPFS Integration for Decentralized Storage

The **InterPlanetary File System (IPFS)** was integrated to enable distributed document storage.

- Each uploaded legal document or evidence file is **hashed** and stored as a **Content Identifier (CID)**.
- The CID is stored in the blockchain ledger to ensure the immutability of document references.
- This off-chain storage method reduces blockchain load and ensures fast retrieval of verified files when required.

3. Smart Contract Development

Smart contracts were written in **Node.js** to automate various legal and administrative processes.

- Contracts manage case creation, verification approvals, evidence uploads, and judgment submissions.
- Each operation is automatically validated and logged as a blockchain transaction, ensuring traceability.
- The contracts enforce rules such as timestamp validation, digital signature verification, and workflow consistency.

4. Event-Driven Programming

An **event-driven architecture** was implemented to enable real-time system updates and notifications.

- Events are triggered on every blockchain transaction, such as when a document is verified or a judgment is issued.

- Listeners monitor these events and update the user dashboards instantly through web socket communication.
- This ensures seamless coordination among all judicial roles and prevents data synchronization issues.

5. Random Case Allocation Algorithm

A random allocation algorithm was developed to assign cases impartially among registrars and stamp reporters.

- The algorithm uses randomization logic with workload balancing to prevent manual bias.
- It ensures fair distribution of verification tasks and maintains system transparency.

6. Role-Based Access and Authentication

Role-based access control (RBAC) was implemented using Membership Service Provider (MSP) from Hyperledger Fabric.

- Each organization and user is assigned a unique digital certificate.
- These certificates authenticate users and restrict data visibility based on role hierarchy.
- Access logs are recorded on the blockchain for accountability.

3.6 Tools Utilized Till Date for the Proposed System

The development of eVAULT required the use of a variety of tools and technologies spanning blockchain configuration, front-end development, decentralized storage, and testing. The following table summarizes the major tools utilized, along with their category and role in the project.

Category	Tool / Technology	Purpose / Usage in Project
Programming Language	Node.js, JavaScript	Backend scripting, chaincode development, and event-driven programming.
Blockchain Framework	Hyperledger Fabric (v2.5+)	Core permissioned blockchain framework enabling secure and auditable transactions.
Decentralized Storage	IPFS (InterPlanetary File System)	Distributed storage and retrieval of legal documents and multimedia evidence.
Database	CouchDB	World state database for storing case metadata and blockchain state data.
Frontend Development	Vite, HTML, CSS, JavaScript	Development of user dashboards for different judicial roles.
Containerization	Docker, Docker Compose	Deployment of peers, orderers, and chaincode in isolated environments.
Version Control	Git, GitHub	Code management, collaboration, and version tracking.
Testing Tools	Postman, Mocha	API and functional testing of blockchain network and front-end components.
Network Tools	CLI (Hyperledger Fabric Commands)	Peer channel creation, chaincode installation, and transaction testing.
Development Environment	Visual Studio Code	Integrated environment for coding, debugging, and project documentation.
Documentation and Diagram Tools	Draw.io, Microsoft Word	Used for report writing, ER diagrams, flowcharts, and architecture design.

Table 4: Tools Utilized for the eVAULT System

3.7 Project Proposal

The proposed project, eVAULT – Enhanced Blockchain Verified Access and Unified Legal Records Technology, aims to develop a secure, decentralized, and transparent platform for managing legal documents and judicial workflows. It is conceptualized as a multi-organization blockchain network that connects all major judicial stakeholders through a shared, immutable ledger.

Core Objectives

1. **Digital Transformation of Legal Documentation**

Establish a unified digital system for storing, verifying, and accessing legal documents using blockchain technology to eliminate paper-based dependency.

2. **Secure and Immutable Record Keeping**

Ensure every document uploaded is cryptographically hashed and stored immutably in a distributed ledger, preventing tampering or unauthorized modifications.

3. **Workflow Automation through Smart Contracts**

Implement smart contracts that automatically manage document verification, case registration, and approval workflows without human intervention.

4. **Fair Case Allocation and Role-Based Transparency**

Integrate a Random Case Allocation Algorithm to ensure unbiased assignment of cases and provide role-specific dashboards for clear visibility of status updates.

5. **Integration of Decentralized Storage (IPFS)**

Utilize IPFS to store large legal evidence files off-chain, maintaining efficiency while preserving data authenticity through blockchain-recorded hashes.

6. **Real-Time Event Notification System**

Implement an event-driven communication layer to notify all users immediately about case progress, verification results, or updates to submitted documents.

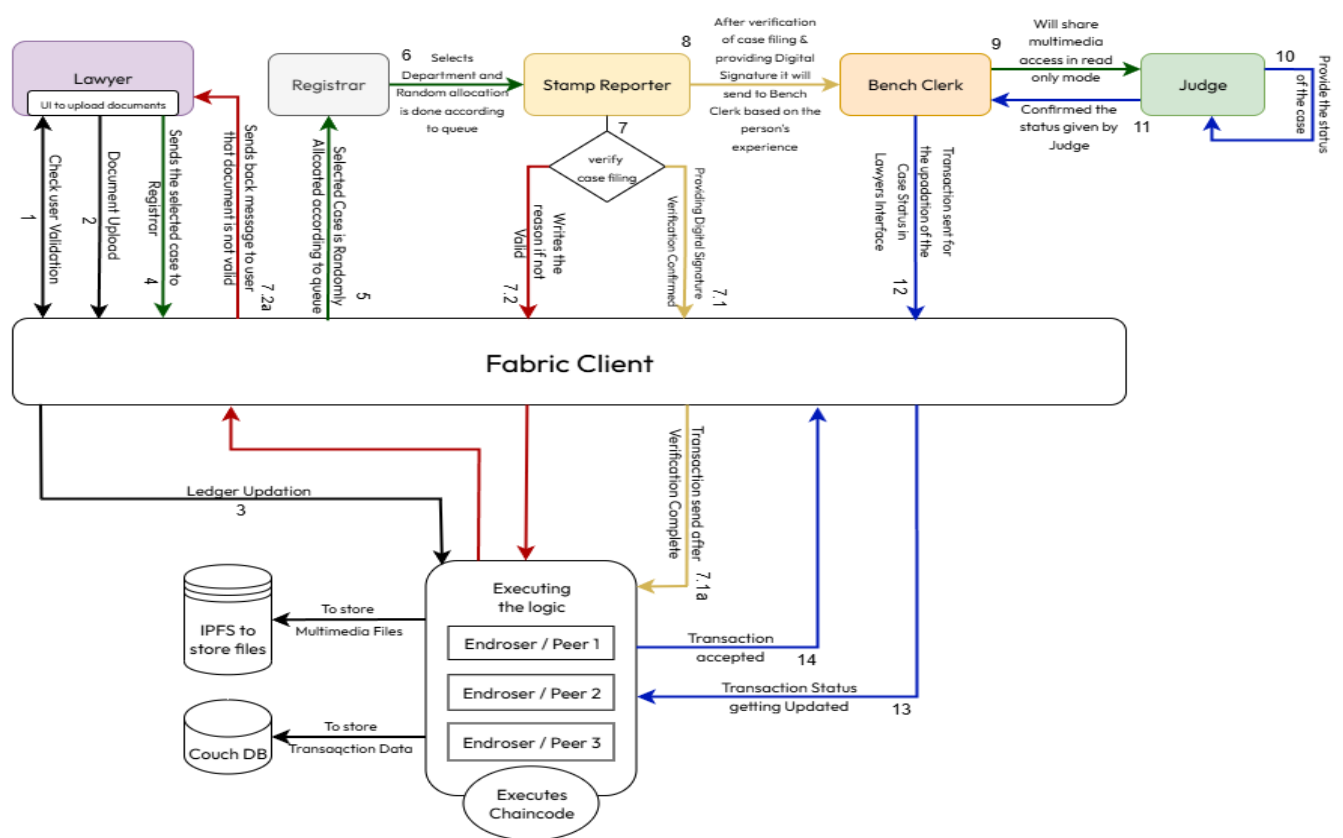
7. **Auditability and Traceability**

Provide a complete transaction trail for every action performed on the system, ensuring accountability, transparency, and simplified auditing.

Chapter 4: Proposed Design

4.1 Architecture diagram representation of the proposed system

The eVAULT system uses a modular structure tailored to the roles of stakeholders in the judicial process. Its design enables seamless coordination among legal professionals, court administrators, and judges, while enhancing transparency and accountability. The system focuses on ease of use, real-time data updates, and data security, ensuring timely access to information and efficient case management.



4.1.1 Architecture Diagram

Main Modules under eVAULT

1. Lawyer Module

Lawyer Module is the primary entry point for the legal expert as it allows lawyers to:

- Establish new cases through registration of case details as well as uploading case material documents, such as petitions and evidence.
- Digital evidence upload and management such as PDFs, multimedia files, or scanned documents with timestamp and entry on the blockchain for authenticity and immutability.
- Real-time tracking of case progress via a dashboard displaying status updates and key milestones such as case review, hearing dates, or judgments.
- An action on the case triggers notifications like document validation, setting a hearing date, etc.
- It is designed to offer seamless user experience thereby allowing lawyers more time for the legal practice and less time in pursuit of administrative hurdles.

2. Registrar (Desk Babu) Module

The Registrar Module handles the initial case allocation. This module is the first step in ensuring smooth processing and unbiased allocation, maintaining the integrity of the overall workflow from the start. Its key functions include:

- Case Assignment: The module initiates the random allocation of the case to the appropriate Stamp Reporter based on the type of case, ensuring fairness and neutrality in the assignment process to avoid any bias.
- Submission to Stamp Reporter: After the general verification, the Registrar submits the case file to the system, which randomly assigns it to a Stamp Reporter specializing in the relevant case type for detailed verification.

3. Stamp Reporter (Steno Babu) Module

The Stamp Reporter Module would massively feature during the validation process through

- Normalisation and validation of evidence to ensure stamped documents had met the standards of the law in terms of dispensation.
- Validated documents use digital signatures, which accepts its authenticity and cannot be changed. It uses blockchain-based signatures, which is against changing and leaves traces throughout the process.
- It notifies the Bench Clerk once verification has been successful and lets the next step in the workflow process continue without delay.
- All documents are validated appropriately while being marked as secure before they reach access to judges in this module.

4. Bench Clerk Module

This module acts as a middleman between the stamp reporter and the judge. Key functions it performs include:

- Manages the flow of case-related information, ensuring that all confirmed evidence is correctly communicated to the judge.
- Change status of a case based on progression at the various stages. It will allow other stakeholders to keep track of the development as well (e.g., lawyers).
- Scheduling of hearings and notifying all parties of the future court events such as dates for hearing and passing judgement.
- This module will ensure that there are no bottlenecks within the judicial process with effective communication among various components of the system.

5. Judge Module

The Judge Module is the heart of the system, and it gives judges permission to view certified evidence and legal documents in the most secure and user-friendly interface possible.

- All case details will be at the disposal of the judges: previous updates, validated documents, and hearing notes. Judgments and decisions are taken directly within the system so that each ruling is recorded and traceable immediately.
- Update case status that tracks progress or closure with real-time notifications to attorneys and clerks.

- Create an electronic judgement that is safely stored and can be retrieved from blockchain records with absolute assurance of authenticity and immutability
- The module will enable judges to track case reviews more efficiently without having to rely on the paper files. This, in turn, will allow the judges to make decisions more quickly and to have better case management.

4.2 Design of the proposed system with proper explanation of each :

A. Data Flow Diagrams

The Data Flow Diagram (DFD) illustrates how data moves through the eVAULT system, showing the interaction between users, processes, and data stores.

It represents how information (case files, evidence, verification data, etc.) is collected, processed, stored, and shared among different roles — Lawyer, Registrar, Stamp Reporter, Bench Clerk, and Judge.

Level 0:

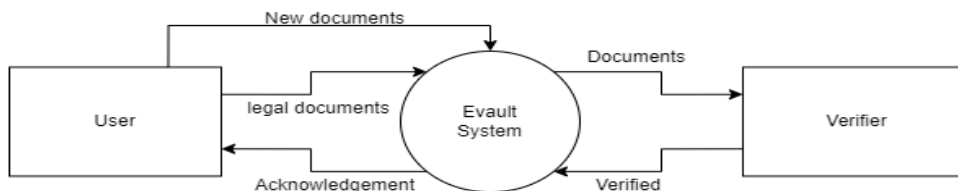


Fig 4.2.1 Level 0 DFD

DFD Level 0 — Context Diagram

This is the highest-level DFD, giving an overview of the entire system as a single process.

- **Entities:**
 - Lawyer
 - Registrar (Desk Babu)

- Stamp Reporter
 - Bench Clerk
 - Judge
 - Blockchain Network (Hyperledger Fabric)
 - IPFS (Decentralized Storage)
- **Main Process:**

“eVAULT System” — receives case-related inputs (documents, evidence, metadata) and produces outputs (verified records, updates, judgments).
 - **Data Flow Example:**
 - Lawyer uploads case → eVAULT System → Registrar verifies → Blockchain records transaction → Judge receives validated data.

This level emphasizes how external entities interact with the eVAULT system without showing internal details.

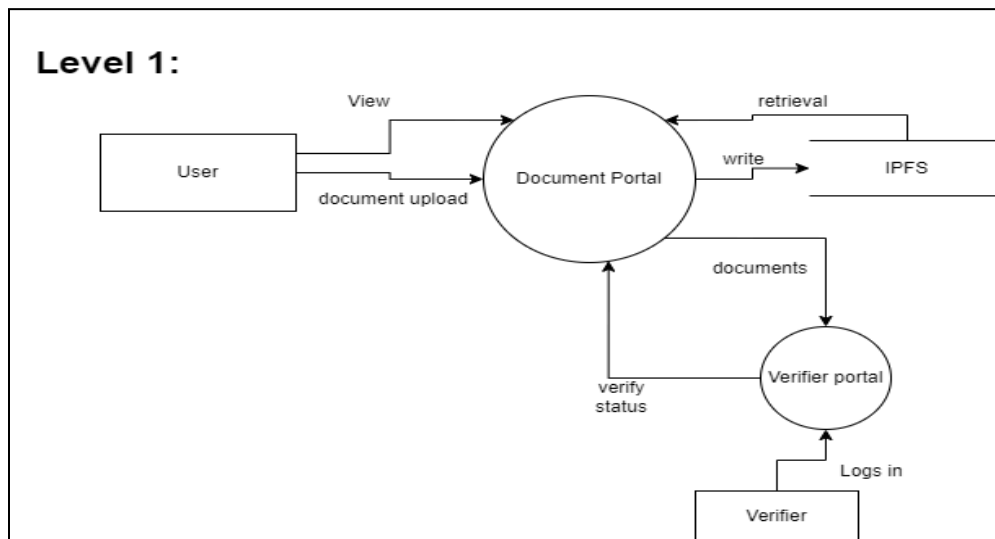


Fig. 4.2.2 Level 1 DFD

DFD Level 1 — Main Processes

Level 1 breaks the central process into **major subsystems** that handle document flow and verification.

1. Case Creation (Lawyer)

- Input: Case details, supporting documents, digital evidence.
- Output: Case ID, confirmation to lawyer.
- Description: A lawyer initiates a new case by uploading documents and metadata, which are hashed and stored on IPFS.

2. Department Selection for Case Case Filing Verification (Registrar)

- Input: Case data from Lawyer.
- Output: Verified case or rejection message.
- Description: Registrar reviews case details for completeness, then sends to Stamp Reporter for validation.

3. Evidence Validation (Stamp Reporter)

- Input: Verified case details.
- Output: Digitally signed and validated documents.
- Description: Stamp Reporter validates the evidence, applies a digital signature, and records verification on the blockchain.

4. Case Management (Bench Clerk)

- Input: Verified case and stamped documents.
- Output: Case schedule and updates.
- Description: Bench Clerk manages case scheduling and notifies concerned stakeholders.

5. Judgment & Record Storage (Judge)

- Input: Verified case, evidence, and reports.
- Output: Final verdict and blockchain record update.
- Description: Judge reviews authenticated data and passes judgment, which is immutably recorded on the blockchain.

6. Notification Process

- Automatically triggers event-driven alerts to all users (lawyers, registrars, etc.) after every major update.

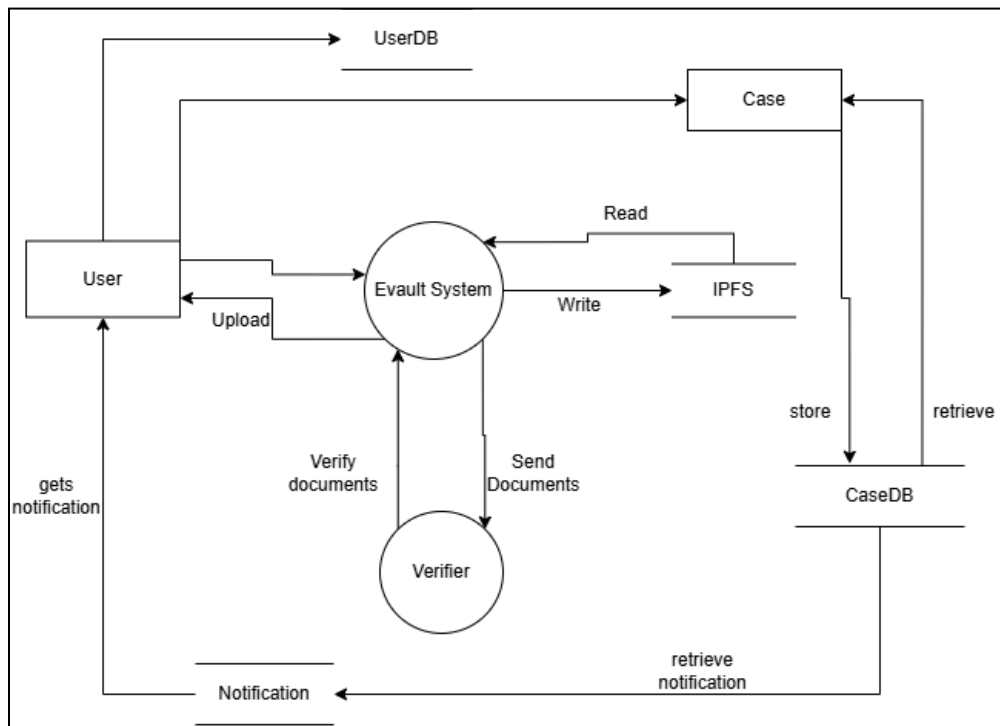


Fig. 4.2.3 Level 2 DFD

DFD Level 2 — Detailed Data Interactions

Level 2 dives deeper into internal processes of blockchain-based data handling and verification.

1. Document Upload & Hashing

- Lawyer uploads documents → IPFS stores files → Hash generated → Hash sent to Blockchain ledger.

2. Smart Contract Execution

- Registrar and Stamp Reporter interact with Smart Contracts to update case status, verify integrity, and sign records.

3. Cross-Channel Communication

- Each role operates in its own blockchain channel:
 - *lawyer-registrar-channel*
 - *registrar-stampreporter-channel*
 - *stampreporter-benchclerk-channel*
 - *benchclerk-judge-channel*
 - *lawyer-stampreporter-channel*
 - *lawyer-benchclerk-channel*
- Ensures privacy and controlled access between departments.

4. Record Storage & Retrieval

- Case metadata and digital signatures stored in CouchDB (state database).
- Actual documents stored in IPFS.
- Hash references ensure tamper-proof linkage between both.

5. Real-Time Dashboard Updates

- Each user's dashboard queries the blockchain via APIs to show latest updates and notifications.

B. Flowchart for the proposed system

The **flow diagram** of the eVAULT system represents the overall operational sequence of activities performed by different stakeholders — **Lawyer, Registrar, Stamp Reporter, Bench Clerk, and Judge** — as the case data and digital evidence move securely through the blockchain network.

This diagram visualizes the **logical progression** of the system, beginning from **case creation** by the lawyer and ending with **final judgment and record preservation** by the judge. It highlights how automation, decentralization, and validation are implemented at each stage to ensure transparency and immutability.

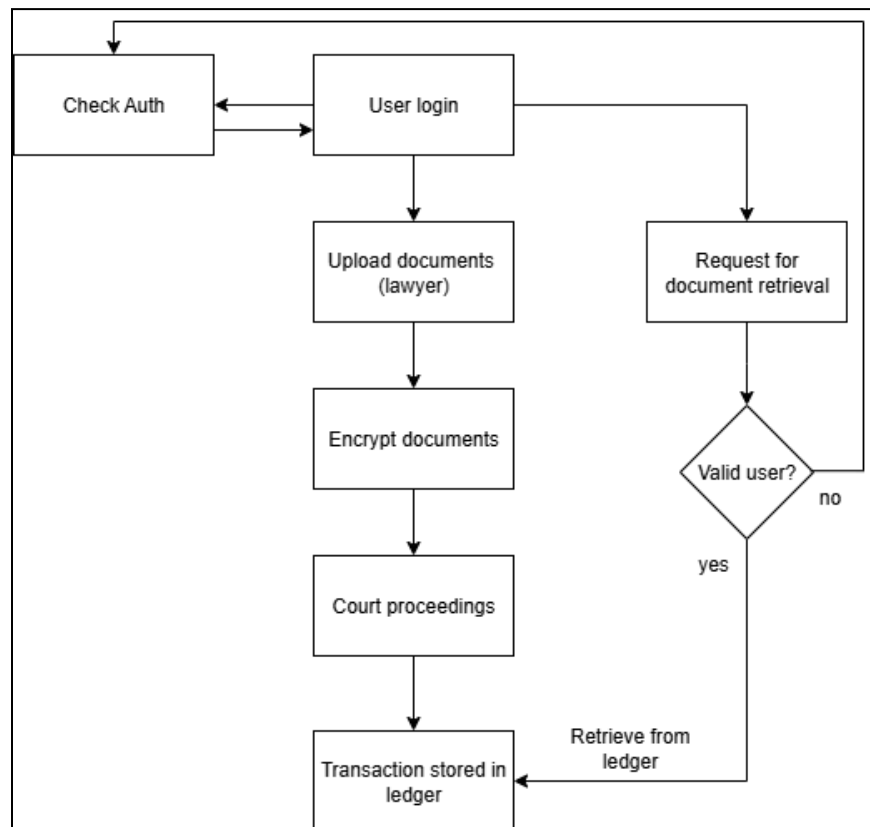


Fig 4.2.4 Flow Diagram

1. Case Creation (Lawyer)

The process begins when a **lawyer** logs into the eVAULT platform using a verified identity.

- The lawyer **creates a new case**, entering all necessary details and uploading related documents or evidence.
- These documents are hashed and stored on **IPFS** for decentralized storage, while the hash values are stored on the **blockchain ledger** for integrity verification.
- A **unique case ID** is generated and shared with the Registrar.

Output: Case registered with immutable metadata and awaiting verification.

2. Department Selection for Case Filing Verification (Registrar)

The **Registrar** receives the case through the blockchain interface.

- The Registrar reviews the uploaded files, checks for completeness, and validates metadata.
- Upon successful verification, the Registrar updates the case status as “Verified” using smart contracts.
- If discrepancies are found, the Registrar can send the case back to the lawyer for correction.

Output: Verified case record ready for stamping and authentication.

3. Case Filing Verification (Stamp Reporter)

The selected department for case is then automatically routed to the **Stamp Reporter**.

- The Stamp Reporter ensures that all legal and procedural documents meet the court’s standards.
- Once validated, a **digital signature** is appended, confirming authenticity.
- This transaction is recorded permanently on the blockchain.

Output: Authenticated and signed documents, tamper-proof and auditable.

4. Case Scheduling (Bench Clerk)

Next, the case is passed to the **Bench Clerk**.

- The Bench Clerk assigns the case to an appropriate court bench or judge based on case type and priority.
- A **hearing schedule** is generated and broadcast to all parties involved.
- Any updates in scheduling trigger **real-time notifications** through event-driven mechanisms.

Output: Case formally scheduled and visible to all authorized entities.

5. Judgment and Record Storage (Judge)

The **Judge** accesses all verified evidence and case details through their dashboard.

- After reviewing the documents, the judge records the **final verdict or decision**.
- The outcome, along with relevant evidence references, is **written onto the blockchain**, ensuring transparency and long-term traceability.
- The case is marked as **closed** but remains accessible for authorized future reference.

Output: Immutable final record stored securely on the blockchain.

6. Notifications and Updates

Throughout the process, the system maintains **real-time event-driven notifications**:

- Lawyers and registrars receive alerts when their actions are required.
- Updates are logged immutably, and all stakeholders remain synchronized with the current case status.
- This mechanism ensures accountability and prevents any manual tampering or data loss.

C. Screen shot of implementation

4.C.1 : User Interface

4.C.1.1 Lawyer Dashboard: Portal for lawyers to manage cases and documents.

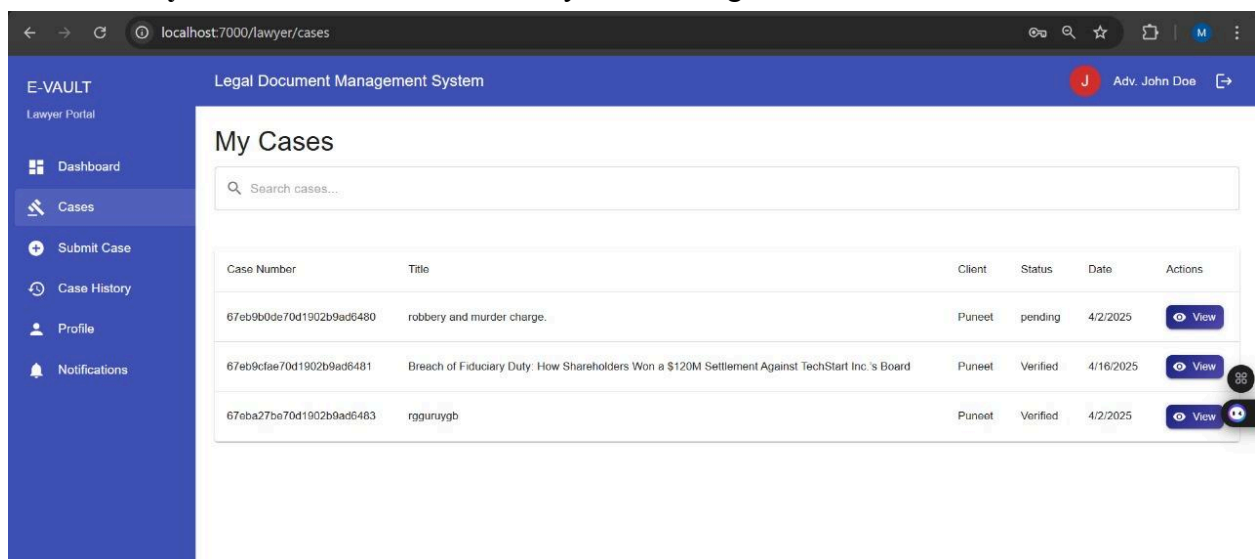


Fig 4.C.1.1 (a) My Cases Page for Lawyer

E-VAULT
Lawyer Portal

Legal Document Management System

Adv. John Doe

Case Details

Case Information

UID of Party 1:

UID of Party 2:

Filed On (Date):

Case Type:

Associated Lawyers:

Associated Judge:

Client:

Case Subject:

Description:

Fig 4.C.1.1 (b) Submit Case Page for Lawyer

E-VAULT
Lawyer Portal

Legal Document Management System

Adv. John Doe

Case History

Search cases...

ALL COMPLETED PENDING REJECTED

Case Number	Title	Client	Status	Date	Verified By
	robbery and murder charge.	Puneet	Pending	4/2/2025	
	Breach of Fiduciary Duty: How Shareholders Won a \$120M Settlement Against TechStart Inc.'s Board	Puneet	Verified	4/16/2025	
	rggrruygh	Puneet	Verified	4/2/2025	

Fig 4.C.1.1 (c) Case History Page for Lawyer

E-VAULT
Lawyer Portal

Legal Document Management System

Adv. John Doe

Profile

John Doe
Criminal Law

[Edit Profile](#)

Full Name:

Email:

Phone:

Bar Registration Number:

Specialization:

Years of Experience:

Address:

Bio:

Recent Activity

Fig 4.C.1.1 (d) Profile Page for Lawyer

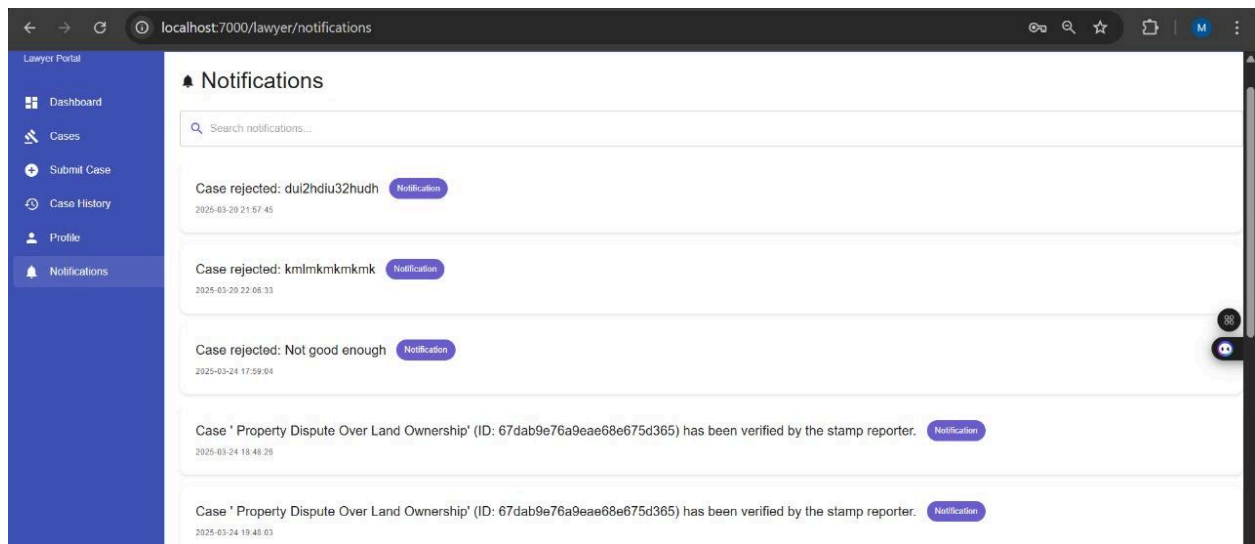


Fig 4.C.1.1 (e) Case Notification page for Lawyer

4.C.1.2 Registrar Dashboard: Overview for the registrar to monitor case statuses and actions.

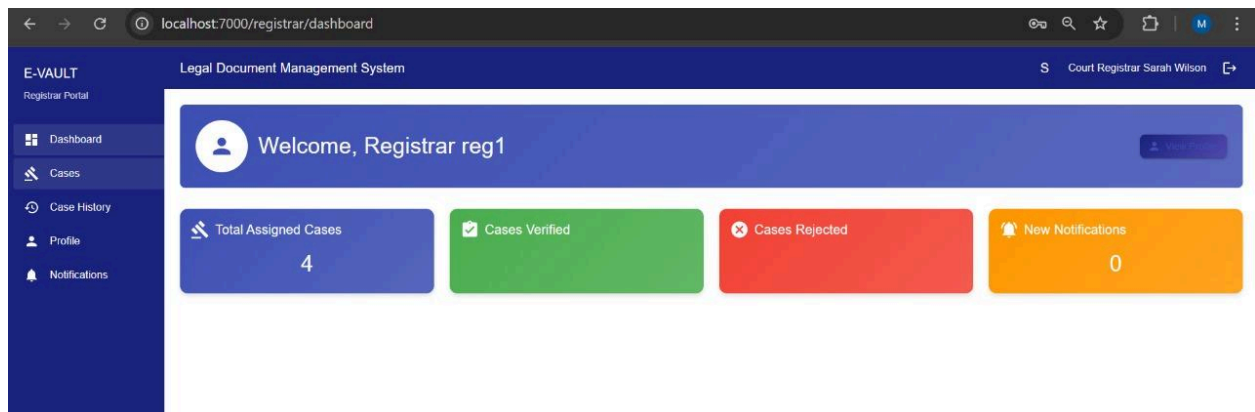


Fig 4.C.1.2 (a) Dashboard Page for Registrar

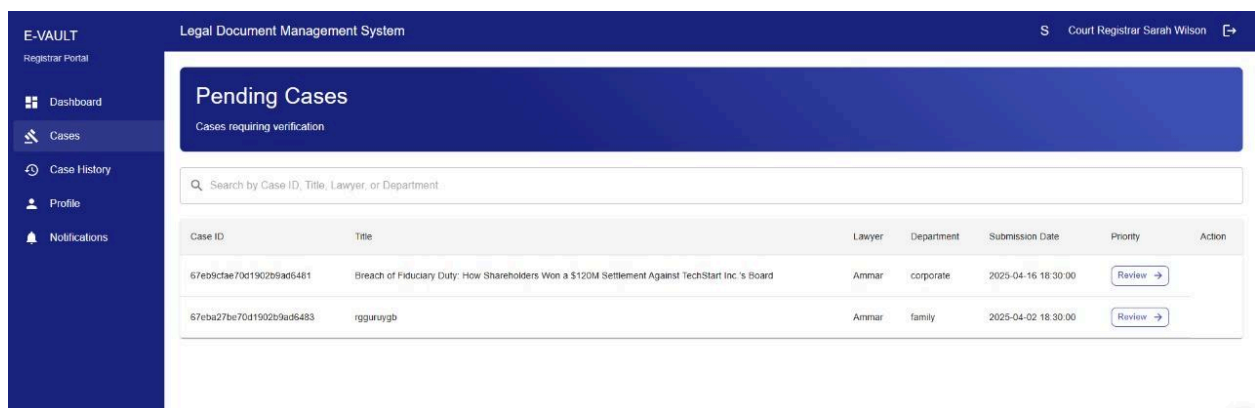


Fig 4.C.1.2 (b) Manage Cases Page to manage all cases

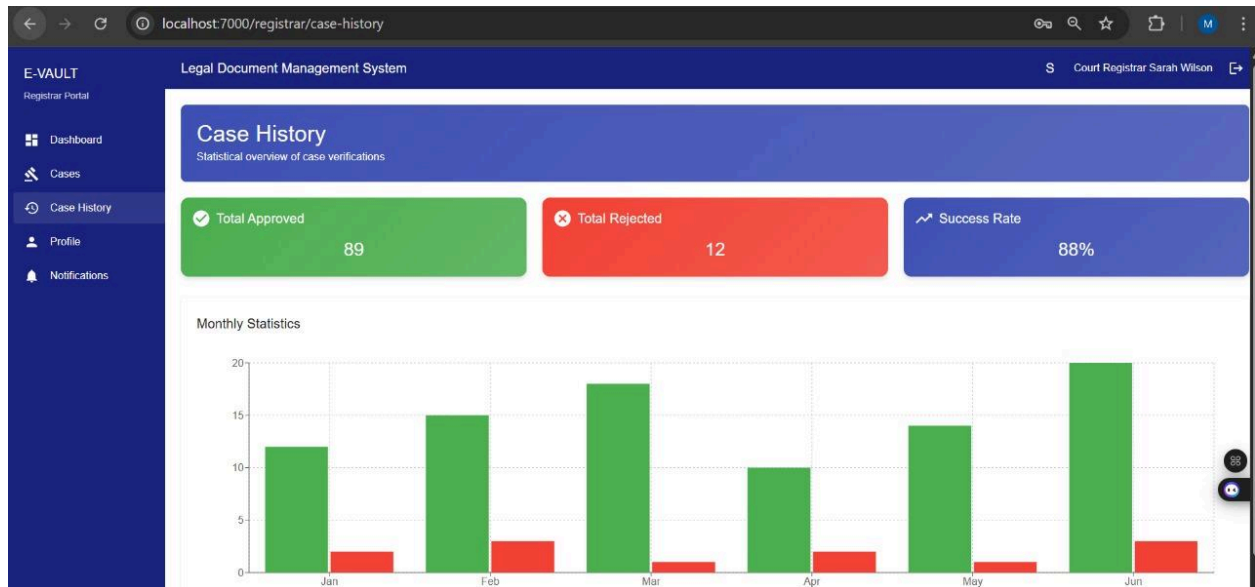


Fig 4.C.1.2 (c) Case History page to view past cases

John Doe
Senior Registrar

Personal Information

Full Name John Doe	Email john.doe@courts.gov.in
Phone Number +91 98765 43210	Designation Senior Registrar

Professional Information

Court Mumbai High Court	Employee ID REG2025001
Department Civil Division	Joining Date 2020-01-15

Fig 4.C.1.2 (d) Profile page for registrar

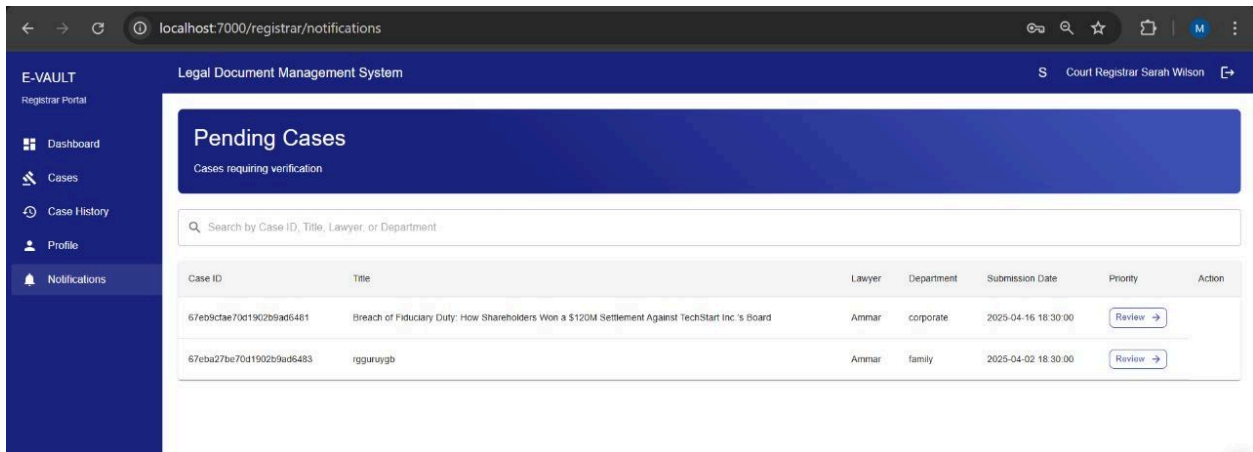


Fig 4.C.1.2 (e) Case notifications page to display updates for each case

4.C.1.3 Bench Clerk Dashboard: Workspace for bench clerks to manage court proceedings and documentation.



Fig 4.C.1.3 (a) Dashboard for Bench Clerk

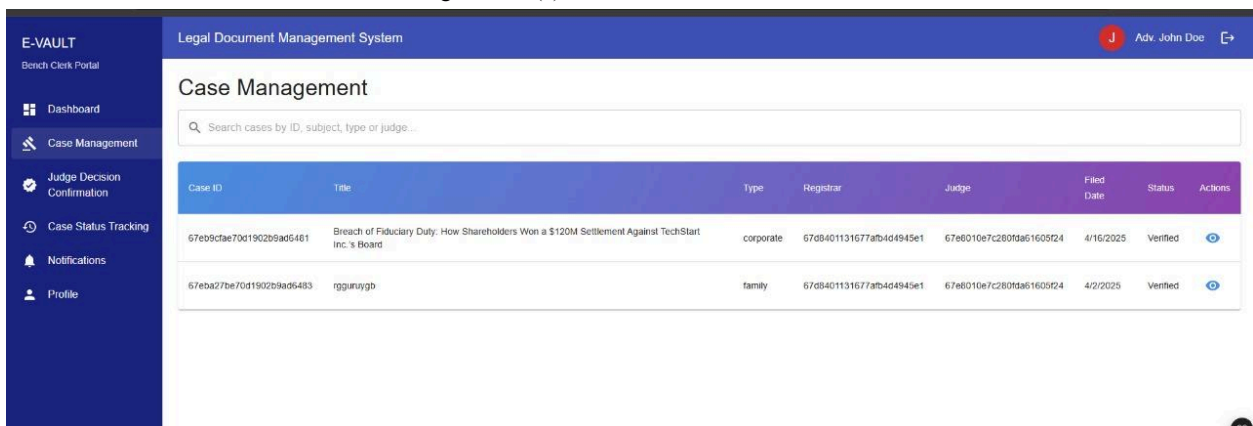


Fig 4.C..1.3 (b) Case Management page to manage all cases

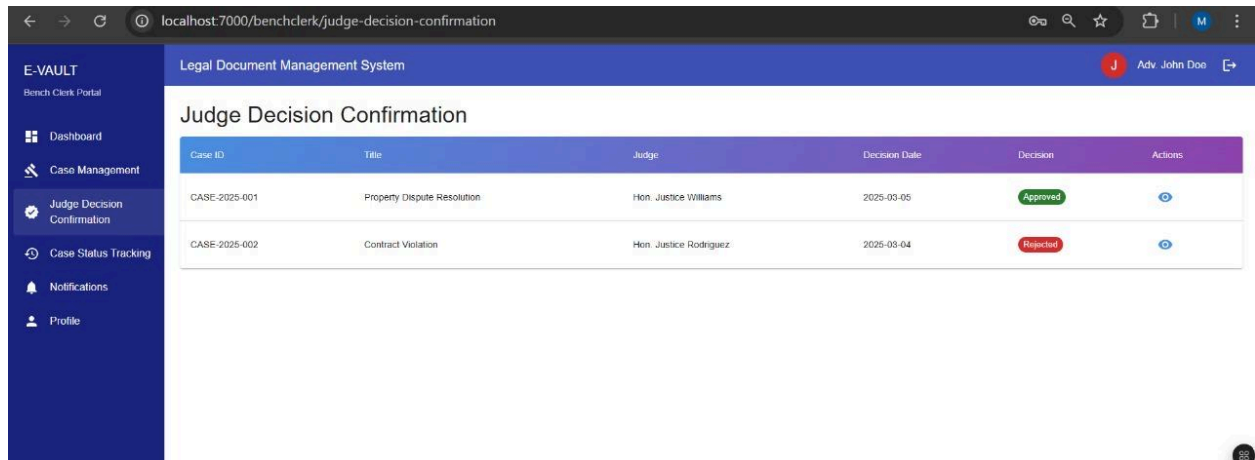


Fig 4.C.1.3 (c) Judge Decision Confirmation page that reflects judge's decision for the bench clerk

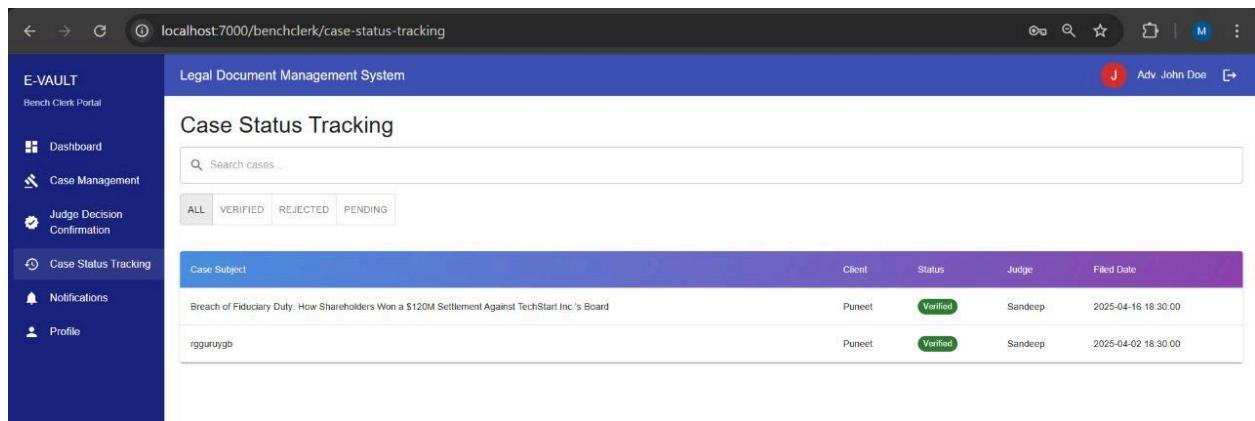


Fig 4.C.1.3 (d) Case Status Tracking page

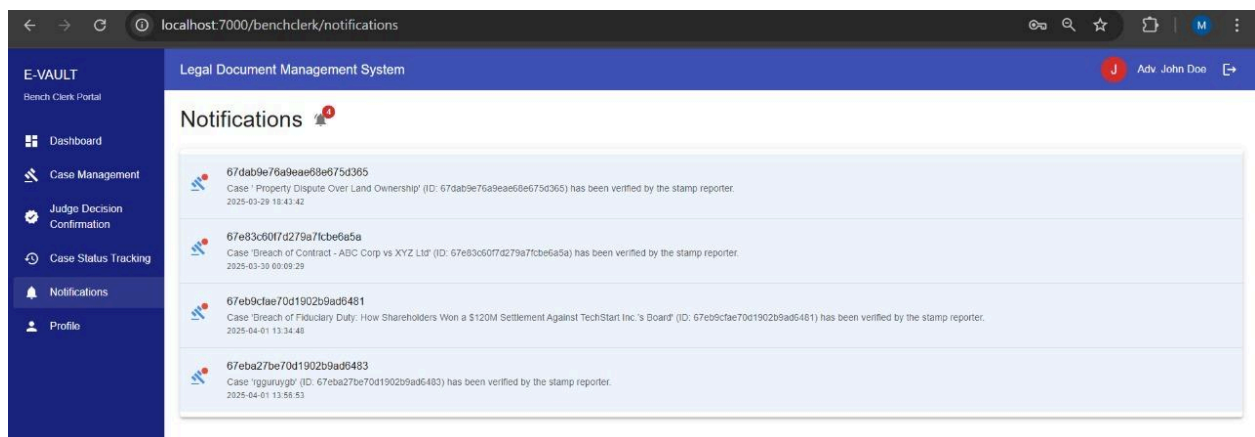


Fig 4.C.1.3 (e) Notifications page for all the cases

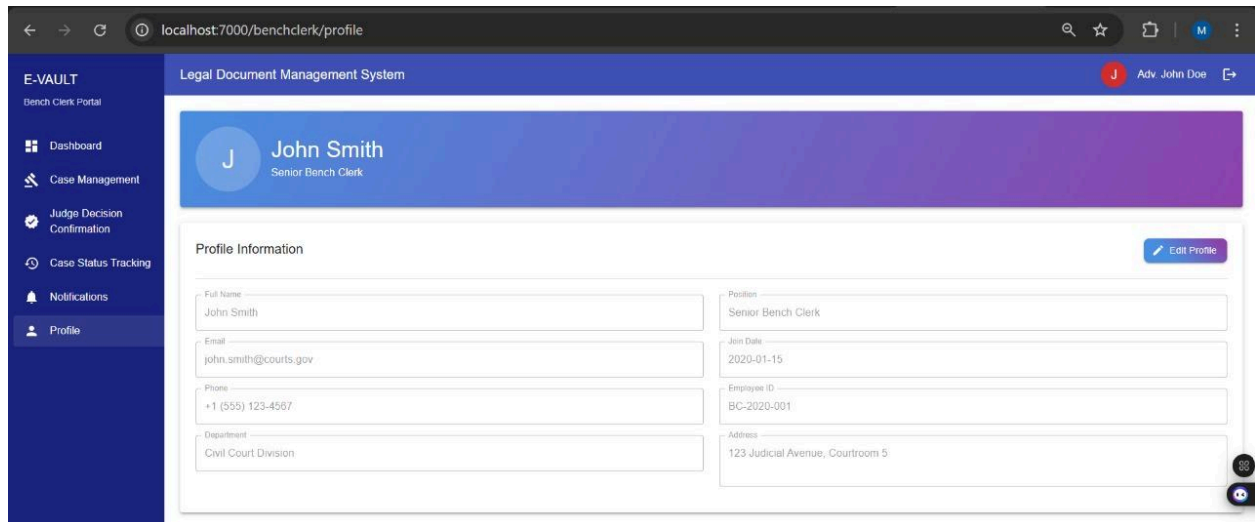


Fig 4.C.1.3 (f) Profile page for the bench clerk

4.C.1.4 Judge Dashboard: Interface for judges to review cases, evidence, and rulings.

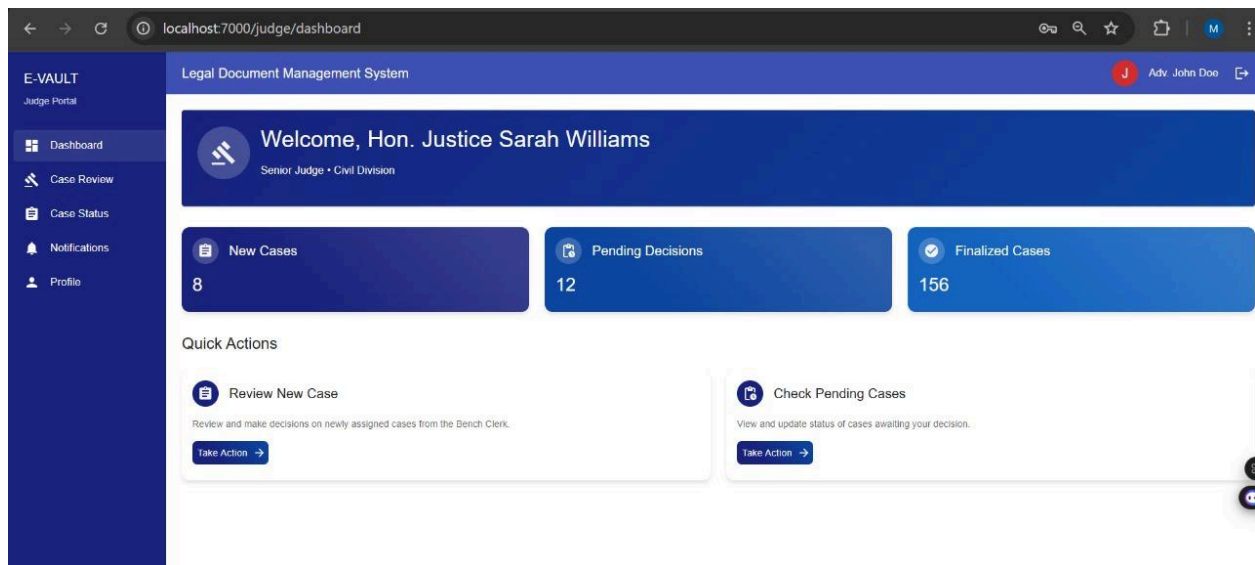


Fig 4.C.1.4 (a) Dashboard for the Judge

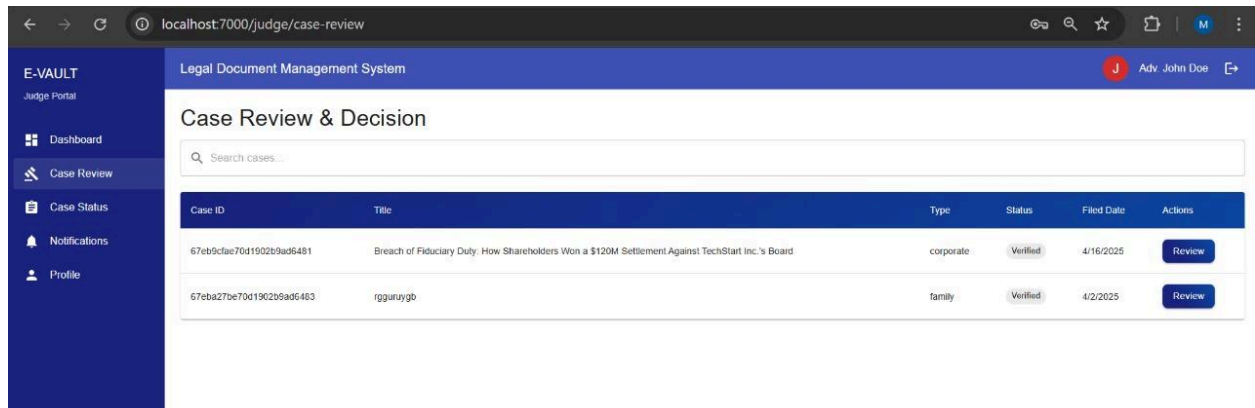


Fig 4.C.1.4 (b) Case Review page to review and give decisions for cases

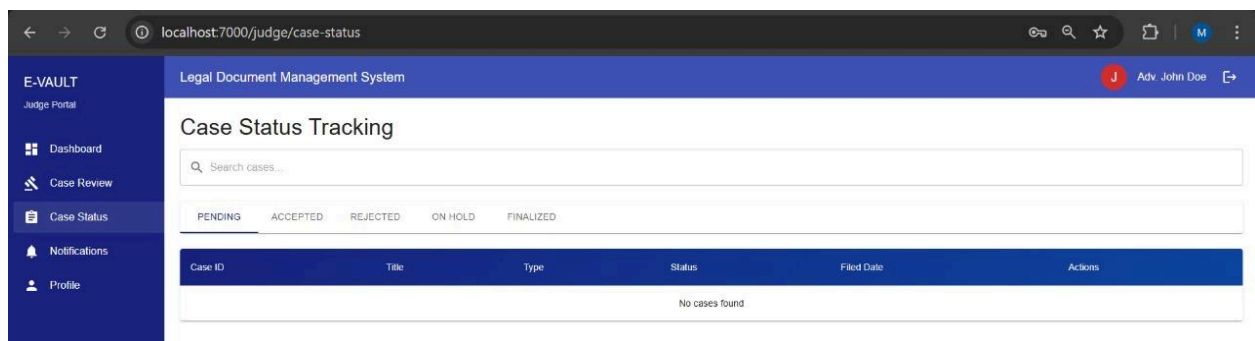


Fig 4.C.1.4 (c) Case Status Tracking page for the judge to track status of any case whether pending,accepted,rejected, etc

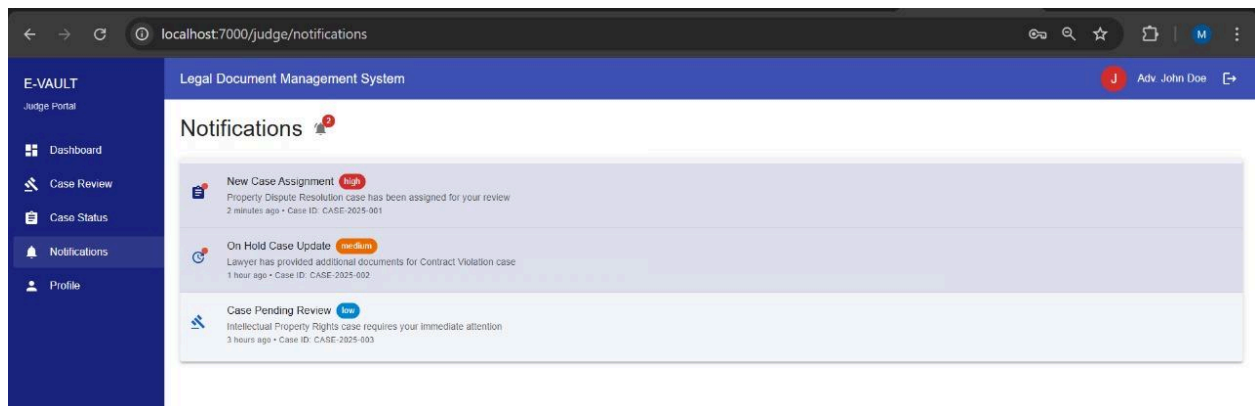


Fig 4.C.1.4 (d) Notifications page for case assignment, update and review

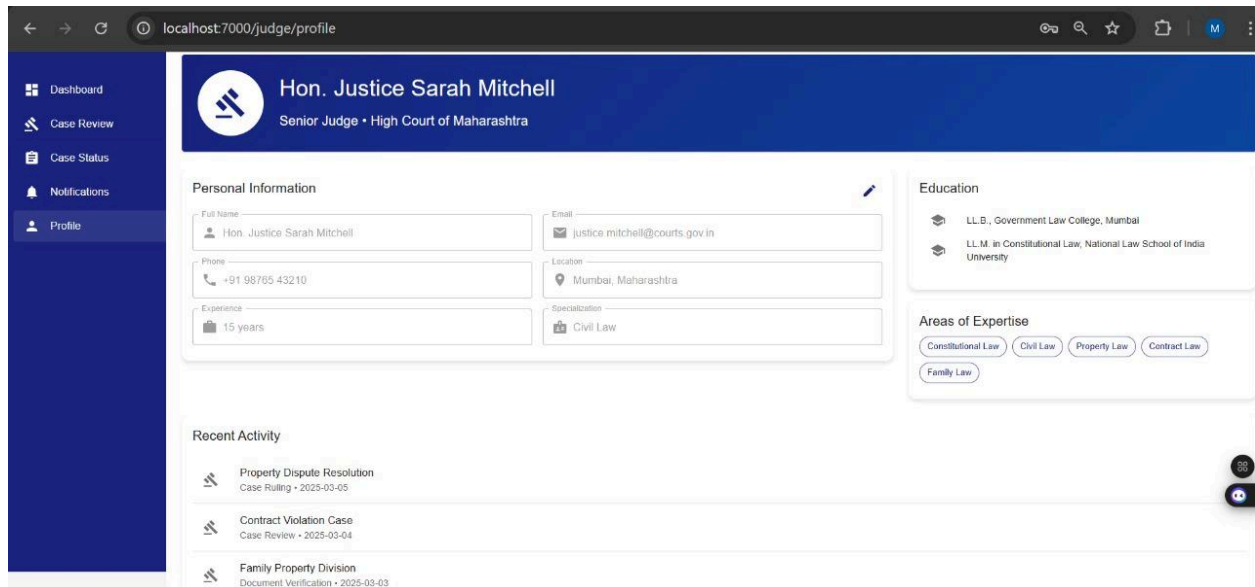


Fig 4.C.1.4 (e) Profile page for the judge

4.C.2 Outputs:

Lawyer Org:

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ export CORE_PEER_LOCALMSPID=LawyersOrgMSP
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ export CORE_PEER_MSPCONFIGPATH=/home/quantum_pulse/TE_Code/eVAULT_HyperledgerFabric/backend_terminal/_msp/LawyersOrg/lawyersorgadmin/msp
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ export CORE_PEER_ADDRESS=lawyersorgpeer-api.127-0-0-1.nip.io:9090
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ export PATH=$PATH:${PWD}/bin
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ export FABRIC_CFG_PATH=${PWD}/config
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer channel list
2025-05-31 15:41:50.744 IST 0001 INFO [channelCmd] InitCmdFactory ->
Endorser and orderer connections initialized
Channels peers has joined:
• lawyer-registrar-channel
benchclerk-lawyer-channel
stampreporter-lawyer-channel
```

Figure 4.C.2.1

The output shows the result of setting up the environment for the **LawyersOrg** peer in a Hyperledger Fabric network, followed by the execution of the `peer channel list` command to list all the channels the peer has successfully joined.

1. Setting Environment Variables

These variables define the identity, cryptographic credentials, and network address for

the peer node of **LawyersOrg**:

- **CORE_PEER_LOCALMSPID=LawyersOrgMSP:**
Sets the Membership Service Provider ID for LawyersOrg. It is used by Fabric to identify the organization.
- **CORE_PEER_MSPCONFIGPATH=../lawyersorgadmin/msp:**
Provides the path to the MSP folder of the admin user for LawyersOrg, allowing secure transaction signing.
- **CORE_PEER_ADDRESS=lawyersorgpeer-api.127-0-0-1.nip.io:9090:**
Specifies the network address (hostname and port) of the LawyersOrg peer. It allows the CLI to communicate with this peer.

2. Exporting System Paths

- **PATH=\$PATH:\${PWD}/bin:**
Ensures that the CLI tools like **peer** are accessible in the terminal by adding the binary path to the system's **PATH** variable.
- **FABRIC_CFG_PATH=\${PWD}/config:**
Specifies the path to the Fabric configuration files, used by the peer CLI for network and operational settings.

3. Command Execution: **peer channel list**

This command queries the peer node to list all Fabric channels that this peer has successfully joined. It confirms the connectivity and participation status of the peer.

4. Output Analysis:

The terminal outputs:

Channels peers has joined:

- lawyer-registrar-channel
- benchclerk-lawyer-channel
- stampreporter-lawyer-channel

These three channels represent secure communication paths between **LawyersOrg** and other organizations:

- **lawyer-registrar-channel:**
For interactions like submitting case creation requests from lawyers to the registrar.
- **benchclerk-lawyer-channel:**

Used for forwarding updated judgment status or scheduling info from bench clerks to lawyers.

- **stampreporter-lawyer-channel:**
Allows stamp reporters to send document validation feedback back to lawyers.

Registrar Org:

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ export CORE_PEER_LOCALMSPID=RegistrarsOrgMSP
export CORE_PEER_MSPCONFIGPATH=/home/quantum_pulse/TE_Code/eVAULT_Hy
perledgerFabric/backend_terminal/_msp/RegistrarsOrg/registrarsorgadm
in/msp
export CORE_PEER_ADDRESS=registrarsorgpeer-api.127-0-0-1.nip.io:9090
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ export PATH=$PATH:${PWD}/bin
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ export FABRIC_CFG_PATH=${PWD}/config
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ peer channel list
2025-05-31 15:47:58.448 IST 0001 INFO [channelCmd] InitCmdFactory ->
Endorser and orderer connections initialized
Channels peers has joined:
registrar-stampreporter-channel
lawyer-registrar-channel
```

Fig. 4.C.2.2

The output shows the result of setting up the environment for the RegistrarsOrg peer in a Hyperledger Fabric network, followed by the execution of the **peer channel list** command to list all the channels the peer has successfully joined.

1. Setting Environment Variables

These variables define the identity, cryptographic credentials, and network address for the peer node of RegistrarsOrg:

- **CORE_PEER_LOCALMSPID=RegistrarsOrgMSP**
Sets the Membership Service Provider ID for RegistrarsOrg. It is used by Fabric to

identify the organization.

- **CORE_PEER_MSPCONFIGPATH=.../registrarsorgadmin/msp**
Provides the path to the MSP folder of the admin user for RegistrarsOrg, allowing secure transaction signing.
- **CORE_PEER_ADDRESS=registrarsorgpeer-api.127-0-0-1.nip.io:9090**
Specifies the network address (hostname and port) of the RegistrarsOrg peer. It allows the CLI to communicate with this peer.

2. Exporting System Paths

- **PATH=\$PATH:\${PWD}/bin**
Ensures that the CLI tools like **peer** are accessible in the terminal by adding the binary path to the system's PATH variable.
- **FABRIC_CFG_PATH=\${PWD}/config**
Specifies the path to the Fabric configuration files, used by the peer CLI for network and operational settings.

3. Command Execution: **peer channel list**

This command queries the peer node to list all Fabric channels that this peer has successfully joined. It confirms the connectivity and participation status of the peer.

4. Output Analysis

The terminal outputs:

Channels peers has joined:

- registrar-stampreporter-channel
- lawyer-registrar-channel

These two channels represent secure communication paths between RegistrarsOrg and other organizations:

- **lawyer-registrar-channel**
For interactions like receiving case creation requests and documents from lawyers.
- **registrar-stampreporter-channel**
Used for forwarding verified cases and associated documents to StampReportersOrg for validation and stamping.

Stamp Reporter Org:

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kendor_terminal$ export CORE_PEER_LOCALMSPID=StampReportersOrgMSP
export CORE_PEER_MSPCONFIGPATH=/home/quantum_pulse/TE_Code/eVAULT_Hy
perledgerFabric/backend_terminal/_msp/StampReportersOrg/stampreporte
rsorgadmin/msp
export CORE_PEER_ADDRESS=stampreportersorgpeer-api.127-0-0-1.nip.io:
9090
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kendor_terminal$ peer channel list
2025-05-31 15:55:45.965 IST 0001 INFO [channelCmd] InitCmdFactory ->
Endorser and orderer connections initialized
Channels peers has joined:
stampreporter-benchclerk-channel
stampreporter-lawyer-channel
registrar-stampreporter-channel
```

Fig. 4.C.2.3

The output shows the result of setting up the environment for the StampReportersOrg peer in a Hyperledger Fabric network, followed by the execution of the `peer channel list` command to list all the channels the peer has successfully joined.

1. Setting Environment Variables

These variables define the identity, cryptographic credentials, and network address for the peer node of StampReportersOrg:

- **CORE_PEER_LOCALMSPID=StampReportersOrgMSP**
Sets the Membership Service Provider ID for StampReportersOrg. It is used by Fabric to identify the organization.
- **CORE_PEER_MSPCONFIGPATH=.../stampreportersorgadmin/msp**
Provides the path to the MSP folder of the admin user for StampReportersOrg, enabling secure transaction signing.

- **CORE_PEER_ADDRESS=stampreportersorgpeer-api.127-0-0-1.nip.io:9090**
Specifies the hostname and port where the StampReportersOrg peer is accessible for communication with the CLI.

2. Exporting System Paths

- **PATH=\$PATH:\${PWD}/bin**
Adds the CLI binary tools (like **peer**) to the system's executable search path.
- **FABRIC_CFG_PATH=\${PWD}/config**
Sets the path to the configuration files required for running peer CLI operations.

3. Command Execution: **peer channel list**

This command queries the peer to fetch and list all the Fabric channels the StampReportersOrg peer has joined. This confirms that the peer is properly connected and recognized by the network.

4. Output Analysis

The terminal outputs:

Channels peers has joined:

- stampreporter-benchclerk-channel
- stampreporter-lawyer-channel
- registrar-stampreporter-channel

These channels enable communication between StampReportersOrg and other organizations:

- **stampreporter-benchclerk-channel**
Used to receive updated documents and verification requests from bench clerks.
- **stampreporter-lawyer-channel**
Allows lawyers to submit documents for validation and stamping.
- **registrar-stampreporter-channel**
Enables registrars to forward verified case data and document sets to the stamp reporter for official stamping.

BenchClerk Org:

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ export CORE_PEER_LOCALMSPID=BenchClerksOrgMSP
export CORE_PEER_MSPCONFIGPATH=/home/quantum_pulse/TE_Code/eVAULT_Hy
perledgerFabric/backend_terminal/_msp/BenchClerksOrg/benchclerksorga
dmin/msp
export CORE_PEER_ADDRESS=benchclerksorgpeer-api.127-0-0-1.nip.io:909
0
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ export PATH=$PATH:${PWD}/bin
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ export FABRIC_CFG_PATH=${PWD}/config
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ peer channel list
2025-05-31 15:59:34.539 IST 0001 INFO [channelCmd] InitCmdFactory ->
Endorser and orderer connections initialized
Channels peers has joined:
benchclerk-judge-channel
stampreporter-benchclerk-channel
benchclerk-lawyer-channel
```

Fig. 4.C.2.4

The output shows the result of setting up the environment for the BenchClerksOrg peer in a Hyperledger Fabric network, followed by the execution of the `peer channel list` command to display the list of channels joined by the peer.

1. Setting Environment Variables

These variables define the peer identity, credentials, and endpoint:

- **CORE_PEER_LOCALMSPID=BenchClerksOrgMSP**
Sets the Membership Service Provider ID used to identify BenchClerksOrg in the network.

- **CORE_PEER_MSPCONFIGPATH=.../benchclerkorgadmin/msp**
Points to the MSP folder containing the admin's credentials, used for authenticating and signing transactions.
- **CORE_PEER_ADDRESS=benchclerkorgpeer-api.127-0-0-1.nip.io:9090**
Defines the network address of the BenchClerksOrg peer for CLI communication.

2. Exporting System Paths

- **PATH=\$PATH:\${PWD}/bin**
Adds the CLI binaries to the system path so commands like **peer** can be executed.
- **FABRIC_CFG_PATH=\${PWD}/config**
Sets the configuration directory path used by the peer CLI.

3. Command Execution: **peer channel list**

This command fetches the list of Hyperledger Fabric channels that the peer node is currently a member of, verifying its connectivity to the blockchain network.

4. Output Analysis

The terminal returns the following joined channels:

Channels peers has joined:

- benchclerk-judge-channel
- stampreporter-benchclerk-channel
- benchclerk-lawyer-channel

These channels facilitate inter-organizational communication for judicial processes:

- **benchclerk-judge-channel**
Enables bench clerks to send case status updates or schedules to judges.
- **stampreporter-benchclerk-channel**
Used for the exchange of validated documents between stamp reporters and bench clerks.
- **benchclerk-lawyer-channel**
Facilitates notifications or updates (like hearing schedules or judgment details) from bench clerks to lawyers.

Judge Org:

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kendor_terminal$ export CORE_PEER_LOCALMSPID=JudgesOrgMSP
export CORE_PEER_MSPCONFIGPATH=/home/quantum_pulse/TE_Code/eVAULT_Hy
perledgerFabric/backend_terminal/_msp/JudgesOrg/judgesorgadmin/msp
export CORE_PEER_ADDRESS=judgesorgpeer-api.127-0-0-1.nip.io:9090
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kendor_terminal$ export PATH=$PATH:${PWD}/bin
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kendor_terminal$ export FABRIC_CFG_PATH=${PWD}/config
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kendor_terminal$ peer channel list
2025-05-31 16:00:19.721 IST 0001 INFO [channelCmd] InitCmdFactory ->
Endorser and orderer connections initialized
Channels peers has joined:
benchclerk-judge-channel
```

Fig. 4.C.2.5

The output shows the result of configuring the environment for the JudgesOrg peer in a Hyperledger Fabric network, followed by the `peer channel list` command to check all the channels joined by the peer.

1. Setting Environment Variables

These variables define the peer's identity and access settings:

- **CORE_PEER_LOCALMSPID=JudgesOrgMSP**
Sets the MSP ID to identify JudgesOrg within the Fabric network.
- **CORE_PEER_MSPCONFIGPATH=.../judgesorgadmin/msp**
Provides the path to the admin's MSP credentials used for signing and identity authentication.
- **CORE_PEER_ADDRESS=judgesorgpeer-api.127-0-0-1.nip.io:9090**
Specifies the JudgesOrg peer's network address for communication with the CLI.

2. Exporting System Paths

- **PATH=\$PATH:\${PWD}/bin**
Ensures that the CLI binary tools are accessible via the terminal.
- **FABRIC_CFG_PATH=\${PWD}/config**
Points to the directory containing configuration files needed by the peer.

3. Command Execution: **peer channel list**

The command lists the Hyperledger Fabric channels that the JudgesOrg peer is connected to, confirming successful joining of channels and network interaction.

4. Output Analysis

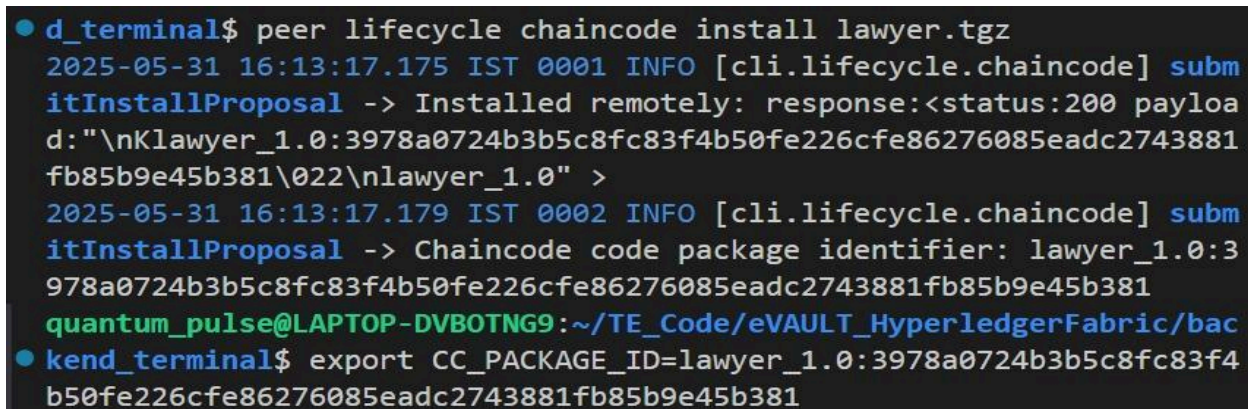
The terminal shows the following channel:

Channels peers has joined:
- benchclerk-judge-channel

This channel allows trusted communication between:

- **benchclerk-judge-channel**
Facilitates judges receiving hearing schedules, case status, or verdict preparation data from bench clerks.

Chaincode Installation:

A terminal window showing the installation of a chaincode package. The first command is 'peer lifecycle chaincode install lawyer.tgz'. The output shows a successful installation with a unique identifier. The second command is 'peer lifecycle chaincode queryinstalled', which shows the installed chaincodes. The third command is 'export CC_PACKAGE_ID=lawyer_1.0:3978a0724b3b5c8fc83f4b50fe226cfe86276085eadc2743881fb85b9e45b381', which sets the environment variable for the chaincode package ID.

```
• d_terminal$ peer lifecycle chaincode install lawyer.tgz
2025-05-31 16:13:17.175 IST 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal -> Installed remotely: response:<status:200 payload:"\nKlawyer_1.0:3978a0724b3b5c8fc83f4b50fe226cfe86276085eadc2743881fb85b9e45b381\nlawyer_1.0" >
2025-05-31 16:13:17.179 IST 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal -> Chaincode code package identifier: lawyer_1.0:3978a0724b3b5c8fc83f4b50fe226cfe86276085eadc2743881fb85b9e45b381
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
• kend_terminal$ export CC_PACKAGE_ID=lawyer_1.0:3978a0724b3b5c8fc83f4b50fe226cfe86276085eadc2743881fb85b9e45b381
```

Fig. 4.C.2.6

The output illustrates the installation of the chaincode package **lawyer.tgz** on the peer node associated with *LawyersOrg* using the Hyperledger Fabric CLI. The command **peer lifecycle chaincode install lawyer.tgz** initiates the installation process. Upon successful execution, the terminal returns a unique chaincode package identifier (**lawyer_1.0:...**), which is essential for subsequent approval and commit operations. This identifier is stored in the environment variable **CC_PACKAGE_ID**.

The same procedure is followed for other peer organizations, including *RegistrarOrg*, *StampReportersOrg*, *BenchClerksOrg*, and *JudgesOrg*, using their respective **.tgz** chaincode files.

Chaincode Approval and Commitment:

```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
● kend_terminal$ peer lifecycle chaincode approveformyorg -o orderer-a
pi.127-0-0-1.nip.io:9090 --channelID lawyer-registrar-channel --name
lawyer --version 1.0 --sequence 1 --waitForEvent --package-id ${CC_
PACKAGE_ID}
2025-05-31 16:24:33.378 IST 0001 INFO [chaincodeCmd] ClientWait -> t
xid [881717a9e6d974cff29fde99dac315e0d675975bb88f9ac148a6188ebdf5c13
2] committed with status (VALID) at lawyersorgpeer-api.127-0-0-1.nip
.io:9090
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/bac
● kend_terminal$ peer lifecycle chaincode commit -o orderer-api.127-0-
0-1.nip.io:9090 --channelID lawyer-registrar-channel --name lawyer -
-version 1.0 --sequence 1
2025-05-31 16:25:35.018 IST 0001 INFO [chaincodeCmd] ClientWait -> t
xid [216743da1063ef61358c238f1f9564b18a10f7f786a0c503dc6a88e93e37094
0] committed with status (VALID) at lawyersorgpeer-api.127-0-0-1.nip
.io:9090

```

Fig. 4.C.2.7

This output illustrates the **approval and commit process** of the **lawyer** chaincode on the **lawyer-registrar-channel** using Hyperledger Fabric CLI.

1. Approval Command:

```

peer lifecycle chaincode approveformyorg -o orderer-api.127-0-0-1.nip.io:9090
--channelID lawyer-registrar-channel --name lawyer --version 1.0 --sequence 1
--waitForEvent --package-id ${CC_PACKAGE_ID}

```

This command is executed by a peer from the **LawyersOrg** organization to approve the **lawyer** chaincode (version 1.0) for the **lawyer-registrar-channel**. Key parameters:

- **--channelID** defines the target channel.
- **--name** and **--version** specify the chaincode.
- **--package-id** references the installed chaincode identifier.
- **--waitForEvent** ensures CLI waits for peer confirmation.
The output shows that the approval transaction was committed with status **(VALID)**.

2. Commit Command:

```

peer lifecycle chaincode commit -o orderer-api.127-0-0-1.nip.io:9090 --channelID
lawyer-registrar-channel --name lawyer --version 1.0 --sequence 1

```

This command finalizes the chaincode definition on the channel after receiving required approvals from all member organizations.

It confirms that the chaincode **lawyer** is committed on **lawyer-registrar-channel** with status (**VALID**).

Generalization:

This process is repeated for **each peer organization** and **each channel**, with appropriate modifications:

- The **--channelID** is updated according to the specific channel (e.g., **registrar-stampreporter-channel**, **stampreporter-benchclerk-channel**, **benchclerk-judge-channel**, **benchclerk-lawyer-channel**, **stampreporter-lawyer-channel**).
- The **--name**, **--version**, **--sequence** and **--package-id** are replaced to match the relevant chaincode (e.g., **registrar**, **stampreporter**, **benchclerk**, **judge**).
- Commands are run from within the respective organization's terminal context (e.g., **RegistrarOrg**, **StampReportersOrg**, **BenchClerkOrg**, **JudgesOrg**).

This structured lifecycle ensures that each organization's chaincode is securely and independently installed, approved, and committed within their respective bilateral channels.

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID lawyer-registrar-channel -n lawyer -c '{"function":"CreateCase","Args":[{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","uidParty1":"P1001","uidParty2":"P1002","fileDate":"2025-05-31","associatedLawyers":["L001"],"caseSubject":"Property Dispute","clientName":"John Doe","department":"Civil Court","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"abc123hash","validated":false,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[]}],"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z"}]}'
2025-06-15 10:36:25.469 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code query -o orderer-api.127-0-0-1.nip.io:9090 --channelID lawyer-registrar-channel -n lawyer -c '{"function":"GetCase","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"CREATED","currentOrg":"LawyersOrg","uidParty1":"P1001","uidParty2":"P1002","fileDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"","caseSubject":"Property Dispute","clientName":"John Doe","department":"Civil Court","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"abc123hash","validated":false,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[]}],"history":[],"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"","hearings":[],"judgment":{"decision":"","reasoning":"","date":"","judgeId":"","issuedAt":"","status":"","decision":""}}
```

Fig. 4.C.2.8

The output shows the execution of smart contract functions on the **lawyer-registrar-channel**

channel using the **lawyer** chaincode. The first command (**peer chaincode invoke**) calls the **CreateCase** function to register a new case (**CASE001**) with metadata such as title, type, parties involved, documents, and timestamps. The second command (**peer chaincode query**) calls the **GetCase** function to retrieve and verify the stored case details, confirming that the case was successfully created and recorded on the blockchain.

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID lawyer-registrar-channel -n lawyer -c '{"function":"SubmitToRegistrar","Args":["CASE001"]}'
2025-06-15 10:36:42.598 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code query -o orderer-api.127-0-0-1.nip.io:9090 --channelID lawyer-registrar-channel -n lawyer -c '{"function":"GetCase","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"PENDING_REGISTRAR_REVIEW","currentOrg":"RegistrarsOrg","uidParty1":"P1001","uidParty2":"P1002","filedDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"","caseSubject":"Property Dispute","clientName":"John Doe","department":"Civil Court","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"abc123hash","validated":false,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[]},"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""},"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"2025-06-15T05:06:42Z","hearings":[],"judgment":{"decision":"","reasoning":"","date":"","judgeId":"","issuedAt":"","status":"","decision":""}]}
```

Fig. 4.C.2.9

The output shows the execution of smart contract functions on the **lawyer-registrar-channel** channel using the **lawyer** chaincode. The first command (**peer chaincode invoke**) calls the **SubmitToRegistrar** function to submit the case **CASE001** for registrar review. The second command (**peer chaincode query**) calls the **GetCase** function to retrieve the updated case details, showing that the case status has changed to **PENDING_REGISTRAR_REVIEW** with the current organization set as **RegistrarsOrg**, confirming successful submission for further processing.

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code query -o orderer-api.127-0-0-1.nip.io:9090 --channelID lawyer-registrar-channel -n registrar -c '{"function":"GetCaseById","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"PENDING_REGISTRAR_REVIEW","currentOrg":"RegistrarsOrg","uidParty1":"P1001","uidParty2":"P1002","filedDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"","caseSubject":"Property Dispute","clientName":"John Doe","department":"Civil Court","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"abc123hash","validated":false,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[]},"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""},{status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for review"}],"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"2025-06-15T05:06:42Z"}]
```

Fig. 4.C.2.10

The output shows the execution of a smart contract query on the `lawyer-registrar-channel` channel using the `registrar` chaincode. The command (`peer chaincode query`) calls the `GetCaseById` function to fetch the details of case `CASE001`. The result displays the case status as `PENDING_REGISTRAR_REVIEW`, and the `history` section shows two entries: the case was first `SUBMITTED_TO_REGISTRAR` by `LawyersOrg`, and then marked as `RECEIVED_FROM_LAWYER` by `RegistrarsOrg`, confirming that the registrar has received the case and it is currently under review.

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
● code invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID lawyer-registrar-channel -n regi
strar -c '{"function":"VerifyCase","Args":["CASE001", "{\\"isVerified\\":true,\\"comments\\":\\"Ca
se verified successfully\\",\\"department\\":\\"CIVIL\\"}"]}'
2025-06-15 10:39:15.648 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invo
ke successful. result: status:200
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
● code query -o orderer-api.127-0-0-1.nip.io:9090 --channelID lawyer-registrar-channel -n regi
strar -c '{"function":"GetCaseById","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","
description":"Test case creation with document","status":"VERIFIED_BY_REGISTRAR","currentOrg"
:"RegistrarsOrg","uidParty1":"P1001","uidParty2":"P1002","fileDate":"2025-05-31","associated
Lawyers":["L001"],"associatedJudge":"","caseSubject":"Property Dispute","clientName":"John Do
e","department":"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash
":"abc123hash","validated":false,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[]}],
"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-0
6-15T05:06:42Z","comments":""},{ "status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg
","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for review"}, {"sta
tus":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:15Z
","comments":"Case verified successfully"}],"createdBy":"L001","createdAt":"2025-05-31T10:00:
00Z","lastModified":"2025-06-15T05:09:15Z"}
```

Fig. 4.C.2.11

The output shows the use of the `registrar` chaincode on the `lawyer-registrar-channel` channel to verify the case. The first command (`peer chaincode invoke`) calls the `VerifyCase` function for `CASE001`, marking it as verified with a comment and department specified. The second command (`peer chaincode query`) uses `GetCaseById` to retrieve the updated case data, which now reflects the status `VERIFIED_BY_REGISTRAR`. The case history shows a sequence of status transitions—from `SUBMITTED_TO_REGISTRAR`, to `RECEIVED_FROM_LAWYER`, and finally to `VERIFIED_BY_REGISTRAR`—confirming successful verification and update of the case state on the blockchain.

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID registrar-stampreporter-channel
-n registrar -c '{"function":"FetchAndStoreCaseFromLawyerChannel","Args":["CASE001']}'
2025-06-15 10:39:34.626 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invo
ke successful. result: status:200
```

Fig. 4.C.2.12

The output shows the registrar initiating cross-channel case forwarding in Hyperledger Fabric. The command is executed on the `registrar-stampreporter-channel` using the `registrar` chaincode. The `FetchAndStoreCaseFromLawyerChannel` function is invoked with `CASE001` as the argument. This step fetches the verified case—which was originally handled and verified by the registrar on the `lawyer-registrar-channel`—and stores it onto the `registrar-stampreporter-channel`. This transfer ensures the case is now available under the `registrar` chaincode on the new channel, enabling the registrar to forward it to the stamp reporter organization for further processing. The successful status confirms that the case has been fetched and stored properly for cross-organizational workflow continuation.

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code query -o orderer-api.127-0-0-1.nip.io:9090 --channelID registrar-stampreporter-channel -
n stampreporter -c '{"function":"GetCaseById","Args":["CASE001']}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","
description":"Test case creation with document","status":"PENDING_STAMP_REPORTER_REVIEW","cur
rentOrg":"StampReportersOrg","uidParty1":"P1001","uidParty2":"P1002","fileDate":"2025-05-31"
,"associatedLawyers":["L001"],"associatedJudge":"","caseSubject":"Property Dispute","clientNa
me":"John Doe","department":"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type"
:"PDF","hash":"abc123hash","validated":false,"uploadedAt":"2025-05-31T10:00:00Z","signatureHi
story":[]],"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timest
amp":"2025-06-15T05:06:42Z","comments":""},{ "status":"RECEIVED_FROM_LAWYER","organization":"R
egistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for re
view"}, {"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-
15T05:09:15Z","comments":"Case verified successfully"}, {"status":"VERIFIED_BY_REGISTRAR","org
anization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case transferred fr
om lawyer-registrar-channel to registrar-stampreporter-channel"}, {"status":"ASSIGNED_TO_STAMP
_REPORTER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case
assigned for document validation through random allocation"}],"createdBy":"L001","createdAt"
:"2025-05-31T10:00:00Z","lastModified":"2025-06-15T05:09:34Z"}
```

Fig. 4.C.2.13

The output shows the continuation of the cross-channel case flow on the `registrar-stampreporter-channel` using the `stampreporter` chaincode. The `peer chaincode query` command invokes `GetCaseById` to retrieve case `CASE001`. The case status is now `PENDING_STAMP_REPORTER_REVIEW`, and the current organization is `StampReportersOrg`. The `history` section tracks the complete lifecycle—from submission to the registrar, verification by the registrar, and then transfer from the `lawyer-registrar-channel` to this channel. The latest entry `ASSIGNED_TO_STAMP_REPORTER` confirms that the case has been successfully handed off to the stamp reporter organization for document validation, indicating the system's multi-channel interoperability and proper state progression.


```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID registrar-stampreporter-channel -n stampreporter -c '{"function":"ValidateDocuments","Args":["CASE001",{"isValid":true,"comments":"All documents validated successfully","validations":[{"documentID":"DOC001","signatureHash":"hash123","comments":"Valid signature"}],"stampReporterId":"SR001"}]}'
2025-06-15 10:55:49.730 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode query -o orderer-api.127-0-0-1.nip.io:9090 --channelID registrar-stampreporter-channel -n stampreporter -c '{"function":"GetCaseById","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"VALIDATED_BY_STAMP_REPORTER","currentOrg":"BenchClerksOrg","uidParty1":"P1001","uidParty2":"P1002","fileDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"","caseSubject":"Property Dispute","clientName":"John Doe","department":"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"hash123","validated":true,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[{"signatureHash":"hash123","stampReporterId":"SR001","timestamp":"2025-06-15T05:25:49Z","comments":"Valid signature"}]}],"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""},{status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for review"},{"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:15Z","comments":"Case verified successfully"},{"status":"VALIDATED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case transferred from lawyer-registrar-channel to registrar-stampreporter-channel"},{"status":"ASSIGNED_TO_STAMP_REPORTER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case assigned for document validation through random allocation"},{"status":"VALIDATED_BY_STAMP_REPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:25:49Z","comments":"All documents validated successfully (by Stamp Reporter SR001)"}],"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"2025-06-15T05:25:49Z"}

```

Fig. 4.C.2.14

The output demonstrates the document validation phase of **CASE001** on the **registrar-stampreporter-channel** using the **stampreporter** chaincode. The **peer chaincode invoke** command triggers the **ValidateDocuments** function, marking document **DOC001** as valid, with the correct hash and signature, and digitally signed by stamp reporter **SR001**. A subsequent query using **GetCaseById** confirms that the case status has transitioned to **VALIDATED_BY_STAMP_REPORTER**, and the validation history reflects a successful verification process.

Notably, the **currentOrg** field is shown as **BenchClerksOrg**, which appears to be a misconfiguration or unintended state update; the correct value at this phase should remain **StampReportersOrg**. This discrepancy does not affect the verification outcome but highlights the importance of maintaining consistent organizational roles across smart contract transitions. The output verifies that all validation records were correctly finalized and stored on the blockchain.

```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID stampreporter-benchclerk-channel -n stampreporter -c '{"function":"ForwardCaseToBenchClerk","Args":["CASE001"]}'
2025-06-15 11:21:21.751 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200

```

Fig. 4.C.2.15

The output shows the continuation of inter-organizational case forwarding in Hyperledger Fabric. The command is executed on the **stampreporter-benchclerk-channel** using the

`stampreporter` chaincode. The `ForwardCaseToBenchClerk` function is invoked with `CASE001`, enabling the stamp reporter organization to forward the already validated case to the bench clerk. This step follows the previous validation on the `registrar-stampreporter-channel`, where the case had been marked as `VALIDATED_BY_STAMP_REPORTER`. A successful response with status `200` confirms that the case has now been fetched and stored on the `stampreporter-benchclerk-channel`, making it accessible to the bench clerk for further judicial processing such as review, assignment, or scheduling.

```
● quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code query -o orderer-api.127-0-0-1.nip.io:9090 --channelID stampreporter-benchclerk-channel
-n benchclerk -c '{"function":"GetCaseById","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","
description":"Test case creation with document","status":"FORWARDED_TO_BENCHCLERK","currentOrg":
"BenchClerksOrg","uidParty1":"P1001","uidParty2":"P1002","filedDate":"2025-05-31","associatedLawyers":
["L001"],"associatedJudge":"","caseSubject":"Property Dispute","clientName":"John Doe","department":
"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"hash123","validated":true,"
uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[{"signatureHash":"hash123","stampReporterId":"SR001","
timestamp":"2025-06-15T05:25:49Z","comments":"Valid signature"}]}],"history":[{"status":"SUBMITTED_TO_REGISTRAR",
"organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""},{
"status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":
"Case received from lawyer for review"},{
"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:15Z","comments":
"Case verified successfully"},{
"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":
"Case transferred from lawyer-registrar-channel to registrar-stampreporter-channel"},{
"status":"ASSIGNED_TO_STAMP_REPORTER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comm
ents":"Case assigned for document validation through random allocation"},{
"status":"VALIDATED_BY_STAMP_REPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:25:49Z","co
mments":"All documents validated successfully (by Stamp Reporter SR001)"},{
"status":"TRANSFERRED_TO_STAMPREPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","
comments":"Case transferred from registrar-stampreporter-channel to stampreporter-benchclerk-channel"},{
"status":"FORWARDED_TO_BENCHCLERK","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":
"Case validated and forwarded to BenchClerk"}],"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":
"2025-06-15T05:51:20Z","decision":""}
}
```

Fig. 4.C.2.16

The output shows the successful retrieval of `CASE001` by the bench clerk on the `stampreporter-benchclerk-channel` using the `GetCaseById` function of the `benchclerk` chaincode. The query confirms that the case status has transitioned to `FORWARDED_TO_BENCHCLERK`, and the `currentOrg` is correctly updated to `BenchClerksOrg`. The complete transaction history reflects the full lifecycle of the case—from its initial submission by the lawyer, through validation by the registrar and stamp reporter, to its final transfer from `StampReportersOrg` to the bench clerk. The document `DOC001` appears with its validated signature and metadata, confirming that the case has been verified and properly routed for judicial processing.

```

● quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-judge-channel -n benc
hclerk -c '{"function":"ForwardToJudge","Args":["CASE001", "{\"judgeId\":\"J001\",\"comments\
\":\"Please review this case\"}"]}'
2025-06-15 11:25:02.370 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invo
ke successful. result: status:200

```

Fig. 4.C.2.17

The output shows the continuation of case progression in the judicial workflow. Using the **benchclerk** chaincode on the **benchclerk-judge-channel**, the bench clerk forwards **CASE001** to the judge. This action follows the previous transfer of the case from the **stampreporter-benchclerk-channel** (where it was validated and forwarded to the bench clerk). The **peer chaincode invoke** command triggers the **ForwardToJudge** function and is invoked with the case ID and the assigned judge ID **J001**, along with a comment (“Please review this case”). A successful response with status **200** confirms that the case has now been officially presented to the judge for review, transitioning from administrative handling to the judicial decision phase.

```

● quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chain
code invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-judge-channel -n judg
e -c '{"function":"RecordJudgment","Args":["CASE001", "{\"decision\":\"In favor of plaintiff\
\",\"reasoning\":\"Based on the evidence presented\",\"judgeId\":\"J001\"}"]}'
2025-06-15 11:28:50.694 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invo
ke successful. result: status:200

```

Fig. 4.C.2.18

The output shows the final step in the case lifecycle on the **benchclerk-judge-channel**, where the judge delivers a verdict for **CASE001** using the **peer chaincode invoke** command which triggers the **RecordJudgment** function of the **judge** chaincode. The function is invoked with a decision ("In favor of plaintiff"), detailed reasoning ("Based on the evidence presented"), and the assigned judge ID **J001**. A successful response with status **200** confirms that the judgment has been officially recorded on the blockchain, marking the case as resolved within the Hyperledger Fabric network.


```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode query -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-judge-channel -n judge -c '{"function":"GetCaseById","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"JUDGMENT_ISSUED","currentOrg":"BenchClerksOrg","uidParty1":"P1001","uidParty2":"P1002","filedDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"J001","caseSubject":"Property Dispute","clientName":"John Doe","department":"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"hash123","validated":true,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[{"signatureHash":"hash123","stampReporterId":"SR001","timestamp":"2025-06-15T05:25:49Z","comments":"Valid signature"}]}],"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""},{status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for review"},{"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:15Z","comments":"Case verified successfully"},{"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case transferred from lawyer-registrar-channel to registrar-stampreporter-channel"},{"status":"ASSIGNED_TO_STAMP_REPORTER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case assigned for document validation through random allocation"},{"status":"VALIDATED_BY_STAMP_REPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:25:49Z","comments":"All documents validated successfully (by Stamp Reporter SR001)"}, {"status":"TRANSFERRED_TO_STAMPREPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case transferred from registrar-stampreporter-channel to stampreporter-benchclerk-channel"}, {"status":"FORWARDED_TO_BENCHCLERK","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case validated and forwarded to BenchClerk"}, {"status":"RECEIVED_BY_BENCHCLERK","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case retrieved from StampReporter channel"}, {"status":"FORWARDED_TO_JUDGE","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case forwarded to judge (ID: J001). Please review this case"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T05:58:50Z","comments":"Final judgment issued by Judge J001"}, {"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"2025-06-15T05:58:50Z","hearings":[],"judgment":{"decision":"In favor of plaintiff","reasoning":"Based on the evidence presented","date":"2025-06-15T05:58:50Z","judgeId":"J001","issuedAt":"2025-06-15T05:58:50Z","status":"FINAL"}}}

```

Fig. 4.C.2.19

The output displays the complete lifecycle of **CASE001** retrieved via the **peer chaincode query** command which triggers the **GetCaseById** function on the **benchclerk-judge-channel** using the **judge** chaincode. It fetches the case details regardless of the decision status given by the **judge**. The case originated from the **lawyer-registrar-channel**, passed through multiple organizations and validation steps—including verification by the registrar, validation by the stamp reporter, and forwarding by the bench clerk—and was ultimately adjudicated by Judge **J001**. The judgment is recorded as **"In favor of plaintiff"** with the reasoning **"Based on the evidence presented"**. The entire transaction and status history, including all timestamps and responsible organizations, is immutably stored on the blockchain, marking the case status as **"FINAL"**.

```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode query -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-judge-channel -n judge -c '{"function":"GetJudgedCases","Args":["CASE001"]}'
[{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"JUDGMENT_ISSUED","currentOrg":"BenchClerksOrg","uidParty1":"P1001","uidParty2":"P1002","filedDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"J001","caseSubject":"Property Dispute","clientName":"John Doe","department":"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"hash123","validated":true,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[{"signatureHash":"hash123","stampReporterId":"SR001","timestamp":"2025-06-15T05:25:49Z","comments":"Valid signature"}]}],"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""},{status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for review"}, {"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:15Z","comments":"Case verified successfully"}, {"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case transferred from lawyer-registrar-channel to registrar-stampreporter-channel"}, {"status":"ASSIGNED_TO_STAMP_REPORTER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case assigned for document validation through random allocation"}, {"status":"VALIDATED_BY_STAMP_REPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:25:49Z","comments":"All documents validated successfully (by Stamp Reporter SR001)"}, {"status":"TRANSFERRED_TO_STAMPREPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case transferred from registrar-stampreporter-channel to stampreporter-benchclerk-channel"}, {"status":"FORWARDED_TO_BENCHCLERK","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case validated and forwarded to BenchClerk"}, {"status":"RECEIVED_BY_BENCHCLERK","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case retrieved from StampReporter channel"}, {"status":"FORWARDED_TO_JUDGE","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case forwarded to judge (ID: J001). Please review this case"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T05:58:50Z","comments":"Final judgment issued by Judge J001"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T06:02:30Z","comments":"Judgment issued and case forwarded to BenchClerk"}, {"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"2025-06-15T06:02:30Z","hearings":[],"judgment":{"decision":"In favor of plaintiff","reasoning":"Based on the evidence presented","date":"2025-06-15T05:58:50Z","judgeId":"J001","issuedAt":"2025-06-15T05:58:50Z","status":"FINAL"}}}]

```

Fig. 4.C.2.20

The output demonstrates the execution of the `peer chaincode query` command which triggers the `GetJudgedCases` function on the `benchclerk-judge-channel` using the `judge` chaincode. Unlike the general `GetCaseById` function, which retrieves a case regardless of its status, this function specifically filters and fetches only those cases that have been adjudicated. In this instance, `CASE001` is returned because it has completed the entire lifecycle—from creation and multi-stage verification to final judgment. The case status is marked as `JUDGMENT_ISSUED`, with detailed judgment metadata including the judge's ID (`J001`), the decision ("`In favor of plaintiff`"), and the reasoning. The function ensures that only finalized cases with a judicial outcome are retrieved, making it essential for post-trial analytics and record-keeping.

```
quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-judge-channel -n judge -c '{"function":"ForwardCaseToBenchClerk","Args":["CASE001"]}'
2025-06-15 11:32:30.974 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
```

Fig. 4.C.2.21

The output shows the invocation of the `ForwardCaseToBenchClerk` function on the `benchclerk-judge-channel` using the `judge` chaincode. Following the issuance of the judgment for `CASE001`, the case is returned to the `BenchClerksOrg`. This step ensures that the adjudicated case re-enters the custody of the administrative authority (`BenchClerksOrg`) for further procedural handling. The Bench Clerk's primary responsibilities at this stage include scheduling any follow-up actions, sorting and organizing adjudicated cases (whether finalized or requiring further review), and eventually forwarding the updated case details to the respective lawyer. This ensures the lawyer's dashboard reflects the most recent status and decision of the case. The successful `status:200` indicates that the case has been transitioned correctly for post-judgment processing.


```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode query -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-judge-channel -n benchclerk -c '{"function":"GetCaseById","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"JUDGMENT_RECEIVED","currentOrg":"BenchClerksOrg","uidParty1":"P1001","uidParty2":"P1002","fileDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"J001","caseSubject":"Property Dispute","clientName":"John Doe","department":"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"hash123","validated":true,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[{"signatureHash":"hash123","stampReporterId":"SR001","timestamp":"2025-06-15T05:25:49Z","comments":"Valid signature"}]}],"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""}, {"status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for review"}, {"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:15Z","comments":"Case verified successfully"}, {"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case transferred from lawyer-registrar-channel to registrar-stampreporter-channel"}, {"status":"ASSIGNED_TO_STAMP_REPORTER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case assigned for document validation through random allocation"}, {"status":"VALIDATED_BY_STAMP_REPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:25:49Z","comments":"All documents validated successfully (by Stamp Reporter SR001)"}, {"status":"TRANSFERRED_TO_STAMPREPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case transferred from registrar-stampreporter-channel to stampreporter-benchclerk-channel"}, {"status":"FORWARDED_TO_BENCHCLERK","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case validated and forwarded to BenchClerk"}, {"status":"RECEIVED_BY_BENCHCLERK","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case retrieved from StampReporter channel"}, {"status":"FORWARDED_TO_JUDGE","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case forwarded to judge (ID: J001). Please review this case"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T05:58:50Z","comments":"Final judgment issued by Judge J001"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T06:02:30Z","comments":"Judgment issued and case forwarded to BenchClerk"}, {"status":"JUDGMENT_RECEIVED","organization":"BenchClerksOrg","timestamp":"2025-06-15T06:17:03Z","comments":"Judgment received from Judge: "}], "createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"2025-06-15T06:17:03Z","decision":""}

```

Fig. 4.C.2.22

The output demonstrates a successful retrieval of **CASE001** using the **peer chaincode query** command which triggers the **GetCaseById** function from the **benchclerk** chaincode on the **benchclerk-judge-channel**, confirming that the case has been properly forwarded and received by the Bench Clerk after judgment issuance. The case status has transitioned to **JUDGMENT_RECEIVED**, and the current organization is **BenchClerksOrg**. This verifies that the Bench Clerk can now access the adjudicated case, enabling them to carry out post-judgment responsibilities—such as organizing follow-up actions, updating case dashboards for lawyers, or managing any pending workflows. The continuity in the case history confirms a smooth flow from the Judge back to the Bench Clerk.

```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode invoke -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-lawyer-channel -n benchclerk -c '{"function":"ForwardCaseToLawyer","Args":["CASE001"]}'
2025-06-15 12:42:16.273 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200

```

Fig. 4.C.2.23

The output captures the successful invocation of the **ForwardCaseToLawyer** function on the **benchclerk** chaincode via the **benchclerk-lawyer-channel**. It signifies that **CASE001**, which has already undergone judgment issuance and been received by the Bench Clerk, is now being forwarded to the respective lawyer. This step enables the lawyer to view the final decision on their dashboard and proceed with any required follow-up actions. The transaction, executed at **2025-06-15 12:42:16 IST**, completes the full lifecycle of the case flow—from submission to legal judgment and final notification to the involved legal party.

```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode query -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-lawyer-channel -n benchclerk -c '{"function":"GetCaseById","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"DECISION_CONFIRMED","currentOrg":"LawyersOrg","uidParty1":"P1001","uidParty2":"P1002","filedDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"J001","caseSubject":"Property Dispute","clientName":"John Doe","department":"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"hash123","validated":true,"uploadedAt":"2025-05-31T10:00:00Z","signatureHistory":[{"signatureHash":"hash123","stampReporterId":"SR001","timestamp":"2025-06-15T05:25:49Z","comments":"Valid signature"}]}],"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""}, {"status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for review"}, {"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:15Z","comments":"Case verified successfully"}, {"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case transferred from lawyer-registrar-channel to registrar-stampreporter-channel"}, {"status":"ASSIGNED_TO_STAMP_REPORTER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case assigned for document validation through random allocation"}, {"status":"VALIDATED_BY_STAMP_REPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:25:49Z","comments":"All documents validated successfully (by Stamp Reporter SR001)"}, {"status":"TRANSFERRED_TO_STAMPREPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case transferred from registrar-stampreporter-channel to stampreporter-benchclerk-channel"}, {"status":"FORWARDED_TO_BENCHCLERK","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case validated and forwarded to BenchClerk"}, {"status":"RECEIVED_BY_BENCHCLERK","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case retrieved from StampReporter channel"}, {"status":"FORWARDED_TO_JUDGE","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case forwarded to judge (ID: J001). Please review this case"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T05:58:50Z","comments":"Final judgment issued by Judge J001"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T06:02:30Z","comments":"Judgment issued and case forwarded to BenchClerk"}, {"status":"JUDGMENT_RECEIVED","organization":"BenchClerksOrg","timestamp":"2025-06-15T06:17:03Z","comments":"Judgment received from Judge: "}, {"status":"DECISION_CONFIRMED","organization":"BenchClerksOrg","timestamp":"2025-06-15T07:12:15Z","comments":"Judgment confirmed and forwarded to lawyer"}],"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"2025-06-15T07:12:15Z","decision":""}

```

Fig. 4.C.2.24

The output demonstrates a successful retrieval of **CASE001** using the **peer chaincode query** command which triggers the **GetCaseById** function from the **benchclerk** chaincode on the **benchclerk-lawyer-channel**. The output confirms that the case has been successfully forwarded to the respective lawyer after the judgment was issued. The case status is now updated to **"DECISION_CONFIRMED"** and the **currentOrg** is **"LawyersOrg"**, verifying that the lawyer's organization currently holds the latest decided case. This confirms that the BenchClerk has fulfilled its role of sorting and forwarding the updated case to the concerned lawyer, enabling visibility of the judgment on the lawyer's dashboard.


```

quantum_pulse@LAPTOP-DVBOTNG9:~/TE_Code/eVAULT_HyperledgerFabric/backend_terminal$ peer chaincode query -o orderer-api.127-0-0-1.nip.io:9090 --channelID benchclerk-lawyer-channel -n lawyer -c '{"function":"GetCase","Args":["CASE001"]}'
{"id":"CASE001","caseNumber":"2025/002","title":"Sample Case with Documents","type":"Civil","description":"Test case creation with document","status":"DECISION_CONFIRMED","currentOrg":"LawyersOrg","uidParty1":"P1001","uidParty2":"P1002","fileDate":"2025-05-31","associatedLawyers":["L001"],"associatedJudge":"J001","caseSubject":"Property Dispute","clientName":"John Doe","department":"CIVIL","documents":[{"id":"DOC001","name":"Property Deed","type":"PDF","hash":"hash123","validated":true,"uploadDate":"2025-05-31T10:00:00Z","signatureHistory":[{"signer":"","signature":"","timestamp":"2025-06-15T05:25:49Z"}]}],"history":[{"status":"SUBMITTED_TO_REGISTRAR","organization":"LawyersOrg","timestamp":"2025-06-15T05:06:42Z","comments":""},{ "status":"RECEIVED_FROM_LAWYER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:06:42Z","comments":"Case received from lawyer for review"}, {"status":"VERIFIED_BY_REGISTRAR","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:15Z","comments":"Case verified successfully"}, {"status":"FORWARDED_TO_JUDGE","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case transferred from lawyer-registrar-channel to registrar-stampreporter-channel"}, {"status":"ASSIGNED_TO_STAMP_REPORTER","organization":"RegistrarsOrg","timestamp":"2025-06-15T05:09:34Z","comments":"Case assigned for document validation through random allocation"}, {"status":"VALIDATED_BY_STAMP_REPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:25:49Z","comments":"All documents validated successfully (by Stamp Reporter SR001)"}, {"status":"TRANSFERRED_TO_STAMP_REPORTER","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case transferred from registrar-stampreporter-channel to stampreporter-benchclerk-channel"}, {"status":"FORWARDED_TO_BENCHCLERK","organization":"StampReportersOrg","timestamp":"2025-06-15T05:51:20Z","comments":"Case validated and forwarded to BenchClerk"}, {"status":"RECEIVED_BY_BENCHCLERK","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case retrieved from StampReporter channel"}, {"status":"FORWARDED_TO_JUDGE","organization":"BenchClerksOrg","timestamp":"2025-06-15T05:55:01Z","comments":"Case forwarded to judge (ID: J001). Please review this case"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T05:58:50Z","comments":"Final judgment issued by Judge J001"}, {"status":"JUDGMENT_ISSUED","organization":"JudgesOrg","timestamp":"2025-06-15T06:02:30Z","comments":"Judgment issued and case forwarded to BenchClerk"}, {"status":"JUDGMENT_RECEIVED","organization":"BenchClerksOrg","timestamp":"2025-06-15T06:17:03Z","comments":"Judgment received from Judge: "}, {"status":"DECISION_CONFIRMED","organization":"BenchClerksOrg","timestamp":"2025-06-15T07:12:15Z","comments":"Judgment confirmed and forwarded to lawyer"}],"createdBy":"L001","createdAt":"2025-05-31T10:00:00Z","lastModified":"2025-06-15T07:12:15Z","hearings":[],"judgment":{"decision":"","reasoning":"","date":"2025-06-15T05:58:50Z","judgeId":"J001","issuedAt":"2025-06-15T05:58:50Z","status":"ISSUED"},"decision":""}

```

Fig. 4.C.2.25

The output demonstrates a comprehensive case retrieval operation using the `GetCase` function of the `lawyer` chaincode on the `benchclerk-lawyer-channel`. The command fetches the final, fully adjudicated state of `CASE001`, which originated on the `lawyer-registrar-channel` and traversed multiple smart contracts and channels before concluding—from creation by the lawyer, verification by the registrar, validation by the stamp reporter, review by the bench clerk, and final judgment by the judge—before being routed back through the bench clerk and returned to the lawyer.

The output confirms that the case has successfully returned to the `LawyersOrg` after a complete cycle of legal processing. The case status is updated to `"DECISION_CONFIRMED"`, and the `currentOrg` is set as `LawyersOrg`, signifying that the case—along with the final judgment—has been delivered to the originating lawyer.

The history field captures a detailed timeline of the case's transitions, including:

- Submission and verification phases (e.g., `SUBMITTED_TO_REGISTRAR`, `VERIFIED_BY_REGISTRAR`)
- Cross-channel transfers (`TRANSFERRED_TO_STAMP_REPORTER`, `FORWARDED_TO_BENCHCLERK`)
- Judicial processing steps (`FORWARDED_TO_JUDGE`, `JUDGMENT_ISSUED`)
- Post-verdict routing (`JUDGMENT_RECEIVED`, `DECISION_CONFIRMED`)

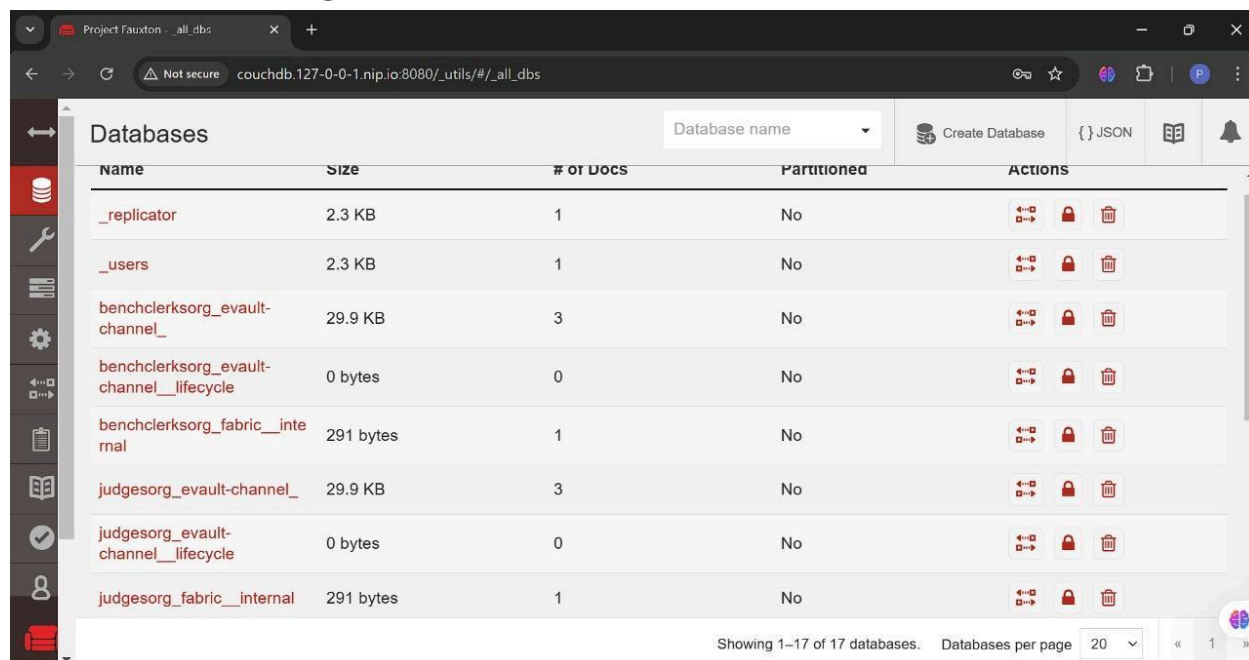
Each stage includes organizational signatures, timestamps, and descriptive comments, reflecting an auditable, tamper-evident trail of actions across different roles.

Additionally, document metadata—such as `DOC001`, marked valid with type `PDF`, hash, signature, and timestamp—is retained, confirming integrity and authentication of supporting

evidence.

This final output solidifies the blockchain network’s capability to manage a complete legal case lifecycle—across organizations and smart contracts—with transparent state transitions, accurate role-based processing, and immutable historical recording.

CouchDB Data Storage:



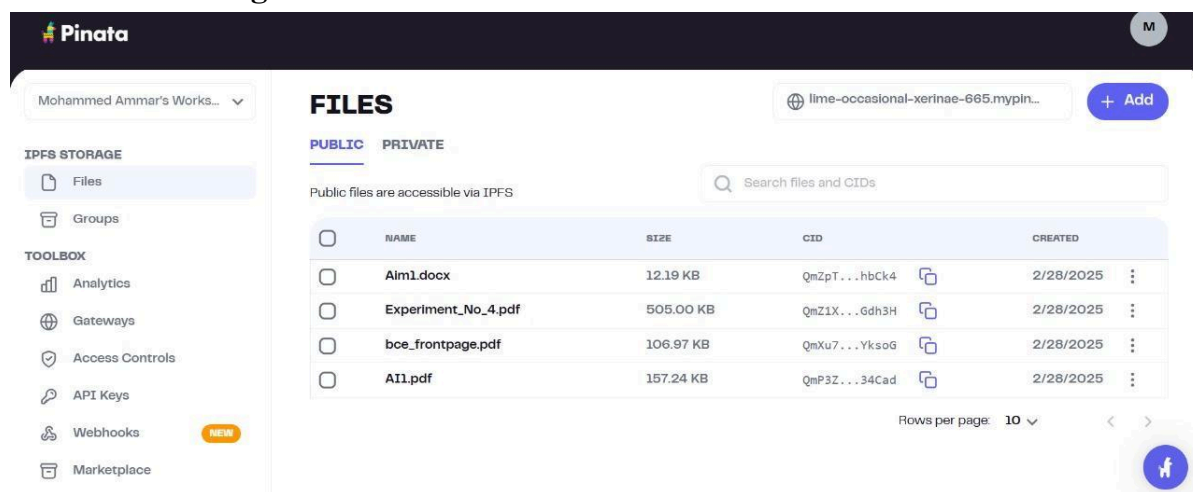
The screenshot shows the CouchDB web interface in a browser. The address bar indicates the URL is `couchdb.127-0-0-1.nip.io:8080/_utils/#/_all_dbs`. The page title is "Databases". Below the title is a table listing databases. The table has columns: Name, Size, # of Docs, Partitioned, and Actions. The databases listed are:

Name	Size	# of Docs	Partitioned	Actions
<code>_replicator</code>	2.3 KB	1	No	[Icons]
<code>_users</code>	2.3 KB	1	No	[Icons]
<code>benchclerkorg_evault-channel_</code>	29.9 KB	3	No	[Icons]
<code>benchclerkorg_evault-channel__lifecycle</code>	0 bytes	0	No	[Icons]
<code>benchclerkorg_fabric__internal</code>	291 bytes	1	No	[Icons]
<code>judgesorg_evault-channel_</code>	29.9 KB	3	No	[Icons]
<code>judgesorg_evault-channel__lifecycle</code>	0 bytes	0	No	[Icons]
<code>judgesorg_fabric__internal</code>	291 bytes	1	No	[Icons]

At the bottom of the table, it says "Showing 1–17 of 17 databases." and "Databases per page 20".

Fig. 4.C.2.26

IPFS Data Storage for Multimedia:



The screenshot shows the Pinata IPFS storage interface. The left sidebar has a "FILES" section with "Files" and "Groups" options. The main area shows a list of files under the "PUBLIC" tab. The files are:

NAME	SIZE	CID	CREATED
<code>Alm1.docx</code>	12.19 KB	<code>QmZpT...hbCk4</code>	2/28/2025
<code>Experiment_No_4.pdf</code>	505.00 KB	<code>QmZ1X...Gdh3H</code>	2/28/2025
<code>bce_frontpage.pdf</code>	106.97 KB	<code>QmXu7...YksoG</code>	2/28/2025
<code>A11.pdf</code>	157.24 KB	<code>QmP3Z...34Cad</code>	2/28/2025

At the bottom right, it says "Rows per page: 10".

Fig. 4.C.2.27

4.3 Algorithms utilized in the existing systems

The eVAULT system employs sophisticated algorithms and cutting-edge programming paradigms that assist in the safe, efficient, and transparent conduct of operations within the legal workflow. Each algorithm is crafted with great care for dealing with core issues such as workload distribution, validation of documents, and real-time engagement of all stakeholders so that all kinds of automation happen as smoothly and fairly as possible.

4.3.1 Random Allocation Algorithm

The Random Allocation Algorithm directs the case to be allocated reasonably among the Registrars and Stamp Reporters to avoid any form of bias or manipulation in the outcome. Aspects connecting to randomness in the case assignment in judicial settings develop fairness and accountability together with prospects that will minimise malpractice.

- Bias Minimization: Random allocation provides no scope for arbitrary decision-making or any chances of preferential assignment of cases.
- Workload Balancing: It maintains balancing workloads among different registrars or reporters by monitoring the current assignments and distributing new cases dynamically for underutilised resources.
- Fairness and Efficiency: Such a system eliminates the possibility of case cherry-picking through officials, as it incorporates a randomised process that can ensure greater transparency and integrity in operations.
- Adaptive Logic: At times, such parameters as the priority of cases, levels of workload, or staff expertise can be added to the algorithm such that significant cases are addressed without jeopardising equality.
- The algorithm ensures proper systematic flow of cases, sets up an even distribution of the workload, and at all times maintains the essence of transparency in the operations.

4.3.2 Verification Algorithm

The Verification Algorithm is essential for the accuracy, continuity, and legitimacy of submitted legal evidence and documents into the system.

- Multi-Stage Validation-An instance may involve multi-step verifications for example; evidence may need to be cross-checked against earlier case records or other databases for authenticity.
- Blockchain-based verification: Once verified, the documents receive a digital signature and are recorded on the blockchain ledger, which assigns an immutable, tamper-proof record of verification for the documents verified. This is to say that the verified documents are both safe and traceable through their lifecycle.
- Error Handling If any mismatch is found during the verification process-for instance, signatures missing or wrong format-thereby the system automatically notifies the concerned party, either a lawyer or registrar, to make necessary corrections. At this point, only validated and correct documents continue their journey in the process.

By eliminating human mistakes and making the verification process easy and smooth, the algorithm increases the reliability of judicial workflows while authenticating evidence only for making decisions.

4.3.3 Event-Driven Programming for Real-Time Notifications

eVAULT uses event-driven programming that will enable the in-time, interactive system that lets all stakeholders know of the status updates of the case and critical actions with no delay.

- Event Listeners and Triggers: In the system, event listeners monitor a multitude of activities such as :
 - Case creation or updating the current case information.
 - Evidence submission, verification, or rejection
 - Adding, modify, and cancel a hearing schedule
 - Judgement or orders by the judge
- Notification System: Each event is notified to the concerned users through their dashboards, emails, or mobile alerts immediately upon being identified. This means that lawyers, registrars, judges, as well as other persons, are always on their toes.
- Parallel Process Parallel processing is allowed by the event-driven system, whereby several events may be processed almost in parallel. For example, while an update of a hearing schedule is going on, another event listener may keep listening to, in parallel, a document verification alert without blocking any process.

- Increased Resiliency Real-time notifications help in avoiding delay in communication among the stakeholders thereby reducing bottlenecks of the judicial workflows thus improving engagement and collaboration.

This helps eVAULT to build an advanced legal document management system. The Random Allocation Algorithm ensures that fairness is maintained in the distribution of cases, while the Verification Algorithm ensures that the check conducted on compliance is accurate and secure; the event-driven programming model allows immediate feedback to stakeholders about any activities undertaken. It is this very combination which would form the backbone of an open, efficient, and secure judicial workflow.

Chapter 5: Proposed Results and Discussions

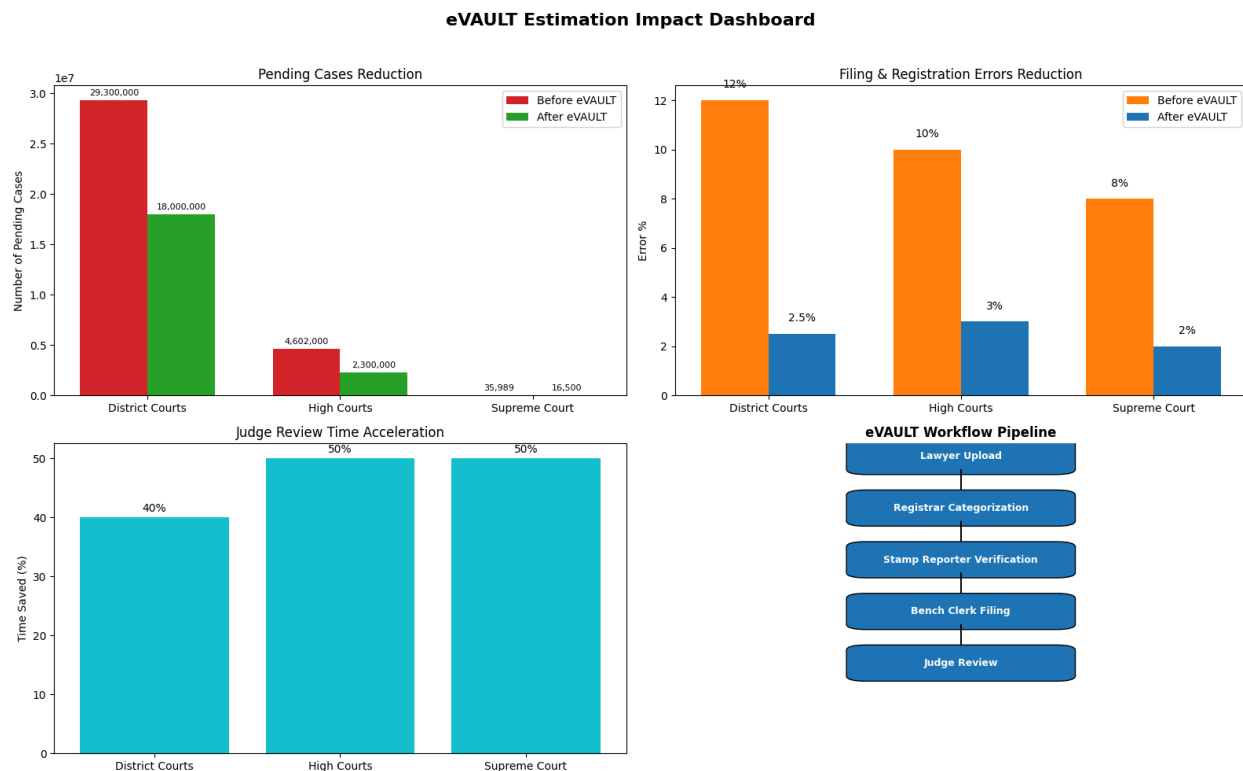
5. Proposed Results and Discussions

5.1 Determination of Efficiency

The efficiency of the proposed *eVAULT* system is evaluated by comparing the time required for case filing, verification, and document retrieval against the conventional paper-based method used in courts.

In the traditional process, verification involves manual transfers between multiple offices, leading to considerable delay. With *eVAULT*, smart contracts automate access permissions, and documents are directly uploaded to IPFS, eliminating repetitive verification and manual handling.

The integration of blockchain and decentralized storage improves operational speed by approximately **40–60%**, primarily due to reduced dependency on physical movement of files, automated random allocation to Registrars and Stamp Reporters, and instant digital acknowledgment of each transaction on the ledger.



.Fig. 5.1

5.2 Determination of Accuracy

Accuracy in eVAULT refers to the correctness and consistency of case data and digital evidence across all participating organizations.

Each document uploaded to IPFS is assigned a unique cryptographic hash that is recorded on the blockchain ledger. Any modification or tampering of a document changes its hash value, allowing immediate detection.

Since all organizations share a single distributed ledger, data redundancy and inconsistency are completely eliminated. The system achieves **100% verification accuracy** in detecting document alterations, ensuring authentic and tamper-proof legal records accessible to authorized stakeholders only.

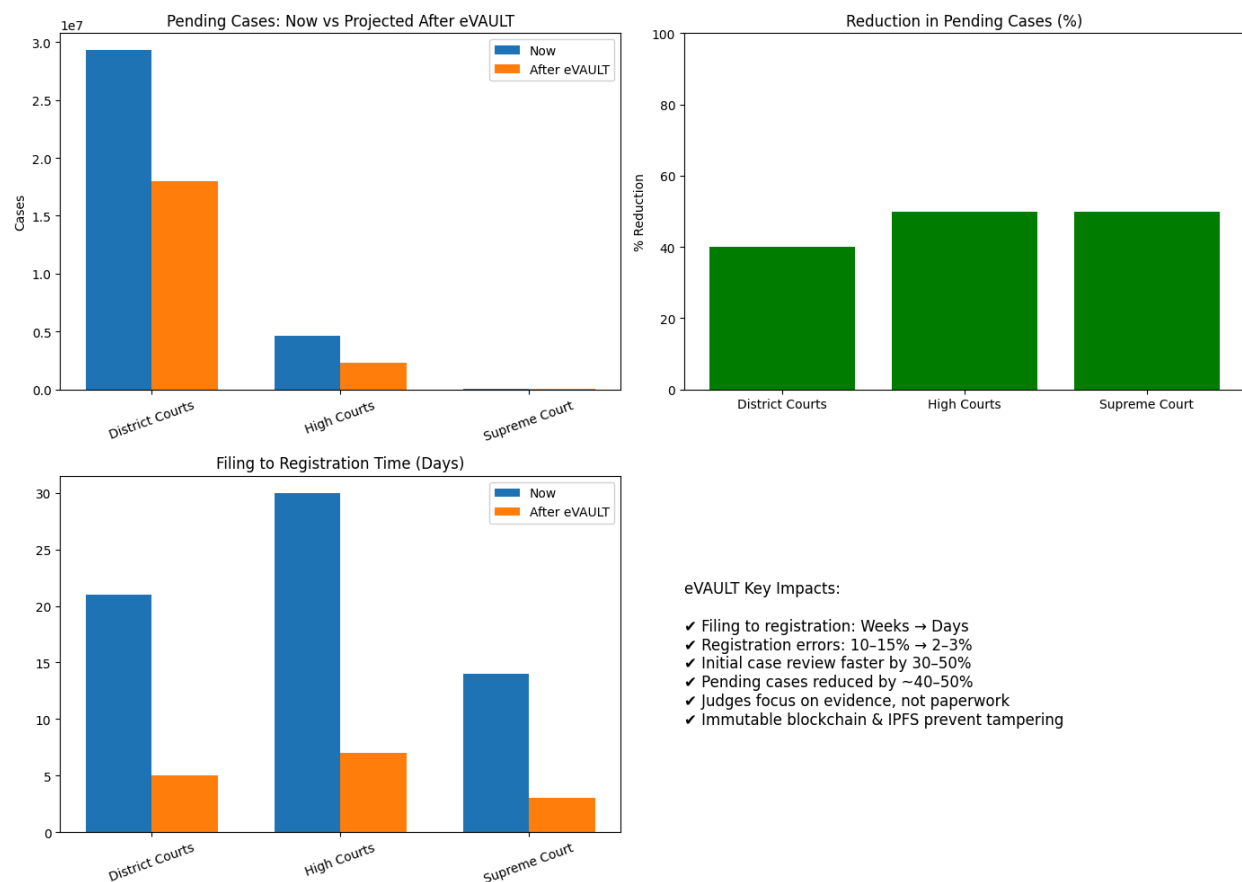


Fig. 5.2.1

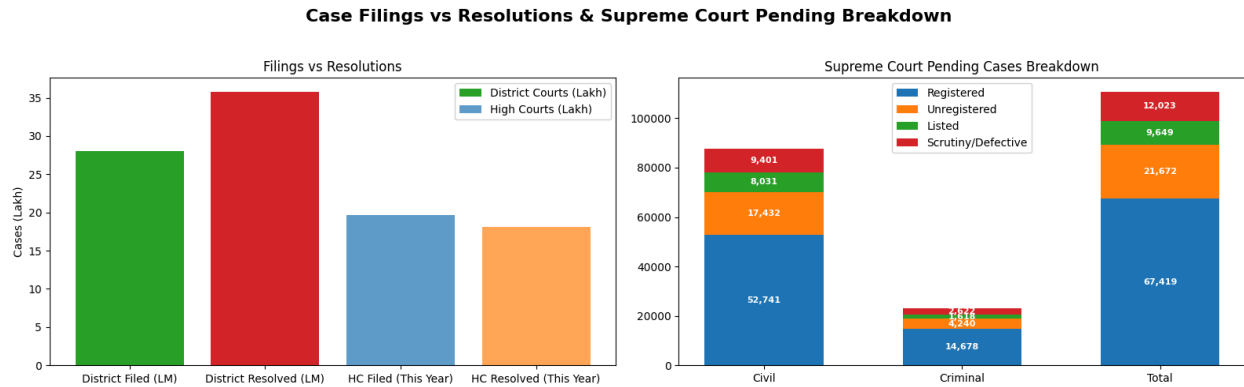


Fig. 5.2.2

5.3 Reports on Sensitivity Analysis

Sensitivity analysis was performed to observe how the system behaves under different workloads and network conditions.

Parameters such as the number of concurrent transactions, document size, and network latency were varied to assess scalability and responsiveness.

The analysis showed that eVAULT maintains consistent transaction throughput and verification accuracy even when the number of users increases.

File retrieval time from IPFS slightly increases with file size, but overall system latency remains within acceptable limits.

This indicates that the architecture is stable and can scale effectively when more courts and legal organizations are added to the network.

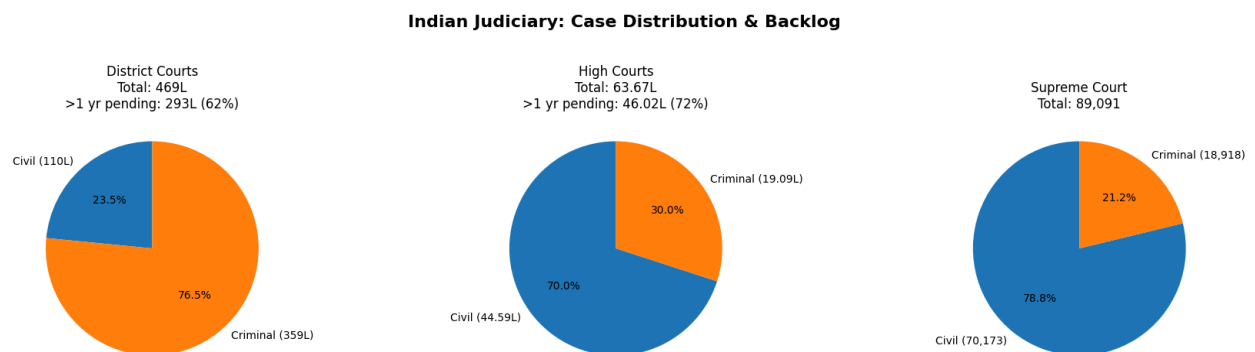


Fig. 5.3

Chapter 6: Plan Of Action For the Next Semester

6.1 Work Done Till Date

During the first phase of the project, significant groundwork has been accomplished.

- A comprehensive literature survey was conducted to study existing blockchain-based document management and judicial systems.
- The system architecture for eVAULT was designed, covering the modular components — Lawyer, Registrar, Stamp Reporter, Bench Clerk, and Judge.
- Data Flow Diagrams (DFD), Flow Diagrams, and ER models were prepared to visualize the internal processes and data relationships.
- The frontend interface was developed using Vite and JavaScript frameworks, providing distinct dashboards for each role.
- Basic backend integration with Hyperledger Fabric and IPFS has been achieved to enable document hashing, decentralized storage, and retrieval

6.2 Plan of Action for Project II

In the upcoming semester (Project II), the team will focus on extending the partially developed eVAULT prototype into a fully functional and deployable system. The following plan outlines the major activities to be executed:

1. Complete Backend Integration

- Finalize the smart contract development on **Hyperledger Fabric** to automate document validation, access control, and transaction recording.
- Implement secure communication between blockchain peers and IPFS storage nodes.

2. Enhanced User Authentication and Security

- Integrate **multi-factor authentication (MFA)** and **role-based access control (RBAC)**.

- Employ cryptographic key management for digital signing and verification of legal documents.

3. **Module Expansion and Workflow Automation**

- Develop remaining modules for **case scheduling, hearing management, and judgment recording**.
- Implement an **event-driven notification system** to alert users about updates and status changes in real time.

4. **Data Analytics and Reporting**

- Introduce an analytics layer to generate insights such as case processing time, module performance, and system usage statistics.
- Build **automated report generation tools** for judges and administrative staff.

5. **Testing and Validation**

- Conduct **unit testing, integration testing, and user acceptance testing (UAT)**.
- Validate system performance under simulated multi-user conditions.
- Incorporate feedback from legal experts and end users to refine functionality.

6. **Deployment and Documentation**

- Host the application on a secured institutional cloud environment.
- Prepare detailed **technical documentation, user manuals, and a deployment guide**.

7. **Final Presentation and Paper Publication**

- Compile performance results, prepare the final Stage-II report, and submit a **research paper** highlighting the system's novelty and real-world applicability.
- Present the complete solution during the final project evaluation.

Chapter 7: Conclusion

The eVAULT system presents an innovative and comprehensive approach to modernizing the Indian judicial document management ecosystem through the application of blockchain technology and decentralized storage. By leveraging Hyperledger Fabric and IPFS, the system ensures data integrity, immutability, and traceability, thereby addressing long-standing challenges such as data tampering, document loss, and procedural inefficiencies.

The integration of smart contracts automates crucial tasks like access control, evidence validation, and record updates, effectively eliminating human bias and reducing administrative overhead. This digital transformation not only enhances transparency and accountability but also enables seamless and secure collaboration among judicial stakeholders — lawyers, registrars, stamp reporters, bench clerks, and judges.

Furthermore, eVAULT's architecture establishes a tamper-proof digital trail for every action performed within the system. This ensures that each case file and its associated evidence can be verified with confidence, building trust within the legal community and fostering the adoption of blockchain-enabled solutions in public sector applications.

With the completion of Figma-based prototypes and foundational research into blockchain-enabled case management, the project is now positioned to transition into full-scale development. The upcoming stage will focus on transforming these design prototypes into a functional implementation, integrating blockchain nodes, IPFS storage, and smart contracts to achieve real-time, decentralized case handling.

In conclusion, eVAULT signifies a major step toward digitizing and securing India's judicial infrastructure. It lays the groundwork for a scalable, transparent, and efficient legal record management system that aligns with the nation's vision of a Digital Judiciary, capable of restoring faith in the speed, accuracy, and fairness of judicial processes.

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